

STATE-OF-THE-ART REVIEW

Skeletal Muscle Dysfunction in Heart Failure

Mechanistic Insights and Therapeutic Implications

Azariyas A. Challa, MD, PhD,^a Ambarish Pandey, MD,^b Peter H. Brubaker, PhD,^c Dalane W. Kitzman, MD,^d Biykem Bozkurt, MD, PhD,^e Dinesh K. Kalra, MD^a

HIGHLIGHTS

- Exercise intolerance remains a major cause of significant QoL limitations in patients with HF.
- HF-associated alterations in SkM structure and metabolism are key drivers of exercise intolerance.
- HF-associated SkM dysfunction is caused by a vicious cycle of oxidative stress, inflammation, proteolysis, and mitochondrial dysfunction.
- Existing HF therapies lack evidence of direct beneficial effects on SkM dysfunction or exercise capacity.
- Novel SkM-targeted therapies are warranted to improve exercise tolerance and QoL of patients with HF.

ABSTRACT

Exercise intolerance is a hallmark of heart failure (HF) and a strong predictor of morbidity and mortality. Accumulating evidence suggests that alterations in skeletal muscle (SkM) structure and metabolism are key contributors to exercise intolerance in patients with HF. These alterations include reduced muscle mass, increased intermuscular adipose tissue, capillary rarefaction, altered fiber-type composition, reduced mitochondrial content, and impaired oxidative phosphorylation. In parallel, an increase in oxidative stress and a decrease in antioxidant defenses, along with derangements in pathways that remove toxic lipid peroxidation products, heighten oxidative stress, perpetuate injury, and establish a vicious cycle of progressive muscle dysfunction. Although exercise training remains the most effective strategy to mitigate SkM dysfunction and improve exercise capacity in patients with HF, novel pharmacotherapies targeting the SkM dysfunction are emerging as potential approaches to treat exercise intolerance in HF. Future studies should delineate the contribution of SkM dysfunction to exercise intolerance in HF, help improve our understanding of its mechanistic underpinnings, and explore the clinical utility of novel SkM-targeted therapies to alleviate exercise intolerance in HF. (JACC Heart Fail. 2026;■:103353) © 2026 by the American College of Cardiology Foundation.

ABBREVIATIONS
AND ACRONYMS**A-VO₂** = arteriovenous oxygen content difference**CPET** = cardiopulmonary exercise test**CR** = caloric restriction**ET** = exercise training**IMAT** = intramuscular adipose tissue**LT** = lactate threshold**NP** = natriuretic peptide**OXPHOS** = oxidative phosphorylation**RAAS** = renin-angiotensin-aldosterone system**ROS** = reactive oxygen species**SkM** = skeletal muscle**VCO₂** = amount of carbon dioxide produced**VE** = minute ventilation**VO₂-peak** = peak oxygen consumption

Heat failure (HF) is a growing global epidemic and is a substantial contributor to morbidity and mortality worldwide. Exercise intolerance, a cardinal symptom of HF, is a key contributor to functional impairment and poor quality of life (QoL) in affected patients and is associated with poor clinical outcomes, including increased mortality.¹⁻⁴ Despite recent advances in pharmacotherapies that reduce hospitalization and mortality, the burden of exercise intolerance and functional impairment among patients with HF remains enormous,³⁻⁷ underscoring the urgent need for novel therapeutic modalities that confer robust and reliable improvements in exercise and functional capacity to patients with HF.^{8,9} The development of novel and effective therapies for exercise intolerance in HF relies on a better understanding of its etiologies and underlying pathophysiologic mechanisms.

Both heart failure with reduced ejection fraction (HFrEF) and heart failure with preserved ejection fraction (HFpEF) are associated with significant skeletal muscle (SkM) abnormalities, including reduced muscle mass, altered fiber-type composition, capillary rarefaction, and reduced mitochondrial content and oxidative capacity.¹⁰⁻¹⁶ Although impaired cardiopulmonary reserve has conventionally been considered central to HF symptomatology, accumulating evidence indicates that alterations in SkM structure and metabolism substantially contribute to exercise intolerance in patients with HF, independent of accompanying cardiorespiratory abnormalities.^{1,7,8,10,14-17} As such, targeting SkM dysfunction is an underappreciated and potentially powerful therapeutic approach for treating exercise intolerance in patients with HF and improving their QoL and functional performance. Importantly, with the rapid growth in the use of incretin-based pharmacotherapies in recent years for obesity and obesity-related conditions, including HF, and the associated

concerns for increased risk of sarcopenia, understanding the complex interaction between SkM dysfunction in HF and the SkM effects of several existing and novel lifestyle, nutritional, and pharmacological interventions to treat exercise intolerance is of paramount importance.

In this review, we provide an overview of the structural and metabolic abnormalities in SkM in HF and their contributions to exercise intolerance compared with cardiopulmonary abnormalities. Importantly, we integrate mechanistic insights into SkM dysfunction in HF with a discussion of therapeutic approaches to alleviate exercise intolerance in patients with HF, including exercise training (ET), nutritional and dietary interventions, and glucagon-like peptide (GLP)-1 receptor agonists (RAs), and highlight recent translational developments, including AdipoRon, apitegromab, bimagrumab BIO101, enobosarm, carnosine, musclin, recombinant human brain-derived neurotrophic factor (BDNF), and Urolithin A. Finally, we discuss current knowledge gaps in the field and priority areas for future studies.

ASSESSING EXERCISE CAPACITY AND SKM
FUNCTION IN HF

Exercise capacity can be objectively quantified by peak oxygen consumption (VO₂-peak) during cardiopulmonary exercise testing (CPET). Other objective measures of exercise capacity include the 6-minute walk distance (6MWD), exercise time to exhaustion (ET), and metabolic equivalents of task from a treadmill test. Among these, CPET is the most reproducible and provides information on patient effort (as reflected in the respiratory exchange ratio) and allows for assessment of the mechanisms limiting exercise capacity. VO₂-peak during a maximal-effort CPET and change in VO₂-peak over time are strong prognostic markers in HF.¹⁸ CPET can be combined with cardiac imaging or invasive hemodynamics to improve its ability to characterize limitations in cardiopulmonary reserve and quantify the contribution of impairments in peripheral O₂

From the ^aDivision of Cardiology, Department of Medicine, University of Louisville School of Medicine, Louisville, Kentucky, USA; ^bDivision of Cardiology, Department of Internal Medicine, University of Texas Southwestern Medical Center, Dallas, Texas, USA; ^cDepartments of Health and Exercise Science, Wake Forest University, Winston-Salem, North Carolina, USA; ^dDepartment of Cardiovascular Medicine, Wake Forest University School of Medicine, Winston-Salem, North Carolina, USA; and the ^eWinters Center for Heart Failure Research, Cardiovascular Research Institute, Baylor College of Medicine, Houston, Texas, USA. The authors attest they are in compliance with human studies committees and animal welfare regulations of the authors' institutions and Food and Drug Administration guidelines, including patient consent where appropriate. For more information, visit the [Author Center](#).

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extraction.¹⁹ The Fick equation states that VO_2 equals cardiac output multiplied by the peripheral O_2 extraction, which is the difference in O_2 content in the arterial blood and mixed venous blood (that is, arteriovenous oxygen difference [A-vO_2]). During peak exercise, cardiac output and peripheral O_2 extraction increase, hence increasing VO_2 -peak. Reduced VO_2 -peak results from reduced cardiac output, reduced A-vO_2 , or both.¹ Ventilatory anaerobic threshold, measured by VO_2 -lactate threshold (LT), is defined as the point at which minute ventilation (VE) increases disproportionately relative to VO_2 due to increased lactate production from anaerobic SkM metabolism.¹⁹ Ventilatory efficiency, measured by the relationship between VE and VCO_2 (amount of carbon dioxide produced), quantified by VE/VCO_2 slope, is associated with cardiac output, pulmonary capillary wedge pressure (PCWP), pulmonary vascular resistance, and respiratory ventilation. Like VO_2 -peak, VO_2 -LT and VE/VCO_2 are also strong prognostic markers in HF.^{20,21}

In addition to the preceding measures of exercise capacity, assessment of SkM strength and SkM mass is particularly relevant to patients with sarcopenia. Isometric handgrip or knee-extension dynamometry and 1-repetition maximum on the bench or leg press are commonly used to assess SkM strength. SkM power or rate of work can be assessed by gait speed, the Timed-Up and Go test, and the 400-m walk test.³ SkM mass can be measured by dual-energy x-ray absorptiometry, bioelectrical impedance analysis, computed tomography, magnetic resonance imaging, ultrasonography, and deuterated creatine dilution.²² However, most of these tools for SkM mass assessment are currently limited to research use and not validated for routine clinical practice. Because of its low cost and lack of ionizing radiation, ultrasonography is emerging as a potentially useful tool for assessing SkM mass.²³ The serum creatinine-cystatin C ratio, a readily accessible biomarker shown to estimate SkM mass in sarcopenic patients, can be useful in screening for sarcopenia but has not yet been validated in patients with HF.²⁴

PERIPHERAL VS CENTRAL MECHANISMS OF EXERCISE INTOLERANCE IN HF

Although impairments in central factors, including cardiac output, stroke volume, chronotropic competence, ventricular compliance, and PCWP, have traditionally been viewed as the predominant mechanisms underlying exercise intolerance in HF,

compelling evidence suggests that peripheral factors, including the peripheral vasculature and SkM structure and function, are major contributors to reduced exercise capacity in HF.¹ In patients with HFpEF, Haykowsky et al²⁵ showed that the difference between rest and peak A-vO_2 was the strongest predictor of VO_2 -peak. Similarly, using invasive hemodynamic measurements and analysis of A-vO_2 during CPET in patients with HFpEF, Dhakal et al²⁶ showed that reduced peak A-vO_2 accounted for >50% of the reduction in VO_2 -peak and is a stronger predictor of exercise capacity than exercise cardiac output, suggesting that peripheral factors affecting O_2 delivery and utilization are the primary drivers of exercise intolerance in HF. In HF pharmacologic trials, which primarily targeted cardiac and/or vascular mechanisms, favorable changes in central hemodynamic parameters, such as PCWP or cardiac output, did not translate into improvements in VO_2 -peak or exercise capacity.²⁷⁻³² Consistently, ET improves VO_2 -peak in patients with HF without significantly altering resting systolic or diastolic cardiac function, peak cardiac output, or stroke volume, underscoring the poor correlation between central cardiac parameters and exercise intolerance.³³⁻³⁶ Indeed, it was shown that 84% of the improvement in VO_2 -peak following 4 months of endurance ET in patients with HFpEF was because of improved A-vO_2 , and not secondary to improvements in cardiac output, indicating the predominant contribution of peripheral factors in the etiology of exercise intolerance in HF.³⁵ Improvements in VO_2 -peak in patients with heart transplantation following ET also supports the notion that SkM dysfunction is a significant contributor to exercise intolerance in HF.³⁷ Furthermore, strong experimental and clinical evidence supports significant independent correlations between abnormalities in SkM structure and metabolism in HF and reduced VO_2 -peak and other measures of exercise capacity, including lactate threshold (VO_2 -LT), exercise time, and 6MWD.^{1,6,11,12,38-41}

Taken together, it is thus evident that the SkM dysfunction is an intrinsic feature of HF, not merely a result of reduced cardiac reserve, and is a predominant contributor to exercise intolerance. This is also consistent with the emerging view of HF as a systemic syndrome with distinct yet interconnected abnormalities in various organ systems, including the heart, vasculature, endothelium, kidneys, autonomic nervous system, adipose tissue, and SkM, among others.

SKM DYSFUNCTION IN HFrEF VS HFpEF

Despite some differences in key pathophysiological pathways and management strategies between HFrEF and HFpEF, similar SkM structural and metabolic abnormalities have been reported in both HF types. SkM from both HFrEF and HFpEF have been shown to exhibit loss of muscle mass, increased intramuscular adipose tissue (IMAT), a switch from type I oxidative to type II glycolytic fibers, capillary rarefaction, reduced mitochondrial content and oxidative capacity, and impaired energy metabolism.^{9,13,40-43} However, there is some evidence that the relative contribution of SkM dysfunction to exercise intolerance may differ between HFpEF and HFrEF. For instance, by using invasive hemodynamics during CPET, Dhakal et al²⁶ demonstrated that patients with HFpEF have a greater limitation in peripheral oxygen extraction than those with HFrEF. Consistently, Weiss et al⁴⁰ reported that subjects with HFpEF deplete SkM phosphocreatine (PCr) more rapidly than those with HFrEF during plantar flexion exercise, suggesting a greater degree of impairment in SkM energy metabolism in HFpEF. Nonetheless, given the substantial overlap in SkM dysfunction between HFrEF and HFpEF, therapies targeting SkM dysfunction may improve exercise capacity across the entire spectrum of left ventricular (LV) ejection fraction in patients with HF.

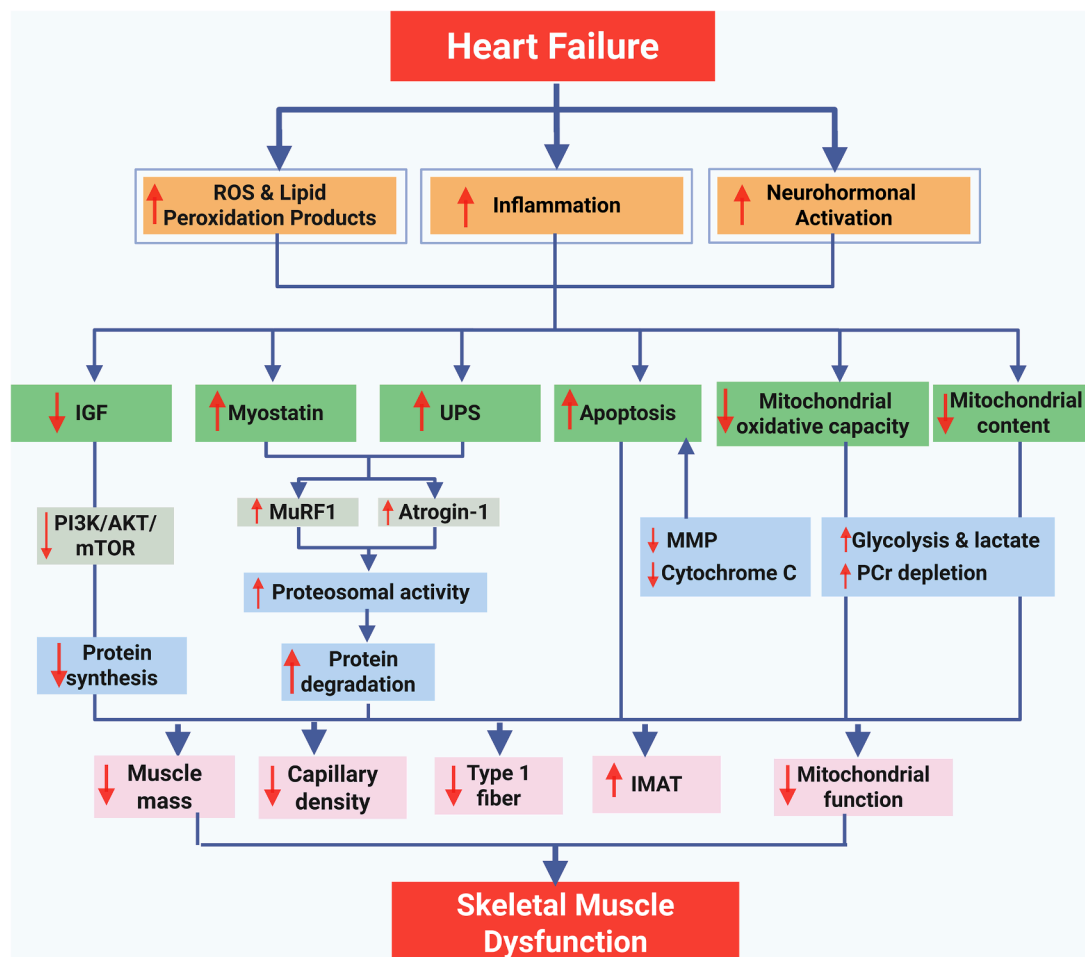
SKM STRUCTURAL REMODELING IN HF

HF is associated with maladaptive changes in SkM structure, including abnormalities in muscle mass, IMAT content, capillary density, and fiber-type composition.^{3,9,14,16,44-46} Reduced muscle mass is a well-documented feature of HF and is an independent determinant of VO₂-peak, functional capacity, and adverse HF outcomes.^{25,47,48} In addition to reduced muscle mass, the force generated by individual myofibers is reduced in HF. Single myofibril studies in patients with HF revealed reduced myosin heavy chain content, resulting in fewer functional myosin-actin cross-bridges and lower maximal Ca²⁺-activated tension.^{49,50} In support of this, Haykowsky et al²⁵ showed that lean body mass correlated with VO₂-peak in patients with HFpEF. In addition, VO₂-peak indexed to lean mass was significantly lower in patients with HFpEF than in healthy control subjects, suggesting that both reduced muscle quantity and quality contribute to the lower VO₂-peak observed in HFpEF.

Evidence from both human and animal studies indicates that the primary mechanism of reduced muscle mass in HF involves an imbalance between protein synthesis and degradation, favoring protein degradation⁵¹ (Figure 1). In both patients with HFrEF and patients with HFpEF, SkM showed decreased insulin-like growth factor (IGF)-1 expression, which stimulates protein synthesis by activating the protein kinase B (Akt)/mechanistic target of rapamycin complex signaling pathway.^{43,52} Consistently, SkM from patients with HF showed reduced phosphorylated Akt levels.⁵³ In a mouse model of HF, SkM-specific overexpression of IGF-1 prevented muscle atrophy by normalizing Akt phosphorylation, suggesting that suppressed IGF-1 activity is a key contributor to reduced muscle mass in HF.⁵⁴ The reduced SkM protein synthesis is accompanied by increased protein degradation via the activated ubiquitin-proteasome system (UPS).⁵¹ SkM from both patients with HFrEF and patients with HFpEF showed increased protein degradation, and elevated levels of muscle ring finger 1 (MuRF1), an ubiquitin-protein E3 ligase of the UPS that targets proteins for proteasomal degradation.^{52,55,56} In addition, myostatin, a catabolic member of the transforming growth factor β family, and atrogin-1, another key E3 ubiquitin ligase involved in UPS-mediated protein degradation, were found to be elevated in SkM of patients with HFpEF.⁴³ Notably, increased UPS-mediated proteolysis correlated with reduced VO₂-peak, suggesting this mechanism of SkM wasting contributes to exercise intolerance in HF.⁵⁵

Other contributors to reduced SkM mass may include increased autophagy, altered mitochondrial dynamics,⁵⁷ myocyte loss via apoptosis,⁵⁸⁻⁶⁰ reduced SkM regenerative capacity,⁶¹⁻⁶³ and increased SkM fat infiltration.⁶⁴ Dysregulated autophagy and imbalance between mitochondrial fusion and fission were associated with SkM wasting in experimental models of HF.^{65,66} Reduced muscle mass in HF is also associated with increased IMAT and IMAT-to-SkM area, both of which are inversely correlated with VO₂-peak.^{64,67,68}

SkM biopsies from patients with HF have shown a decreased capillary-to-fiber ratio associated with reduced VO₂-peak.^{11,64,69} In addition to capillary rarefaction, HF is associated with endothelial dysfunction, which contributes to exercise intolerance.^{70,71} Compared with patients with hypertension, patients with HFpEF had lower cardiac output, more symptoms at low workloads, and blunted exercise-induced peripheral vasodilation, each of which correlated with VO₂-peak in patients with HFpEF. In an observational study involving both HFpEF and

FIGURE 1 Pathophysiology of Skeletal Muscle Dysfunction in Heart Failure

Heart failure is associated with significant structural and metabolic skeletal muscle abnormalities, including reduced muscle mass, altered fiber-type composition, capillary rarefaction, increased intermuscular adipose tissue, and reduced mitochondrial content and oxidative capacity. Heart failure-associated oxidative stress, inflammation, and neurohormonal activation trigger an imbalance between protein degradation and synthesis, apoptosis, and mitochondrial dysfunction. Protein degradation in heart failure involves increased myostatin activity, targeting enzymes of the UPS (ubiquitin-protein E3 ligases, MuRF1, and atrogin-1), and the proteasome. Protein synthesis via IGF signaling is suppressed. Note that mitochondrial dysfunction contributes to the switch in fiber type toward a more glycolytic phenotype, further increases ROS production, and can trigger apoptotic pathways, thereby forming a vicious cycle culminating in skeletal muscle dysfunction and exercise intolerance. IGF = insulin-like growth factor; IMAT = intermuscular adipose tissue; MMP = mitochondrial membrane potential; mTOR = mechanistic target of rapamycin; MuRF1 = muscle ring finger 1; PCr = phosphocreatine; PI3K/Akt = phosphoinositide 3-kinase/protein kinase B; ROS = reactive oxygen species; UPS = ubiquitin-proteasome system.

HFrEF, impaired endothelial function was associated with loss of SkM mass and reduced VO₂-peak.⁷² SkM microvascular dysfunction in HF is likely the result of systemic inflammation, oxidative stress, and excessive neurohormonal activation stemming from HF and associated comorbidities.^{73,74}

SkM is composed of type I (slow-twitch) fibers and type II (fast-twitch) fibers. Type I fibers have higher mitochondrial density and rely on oxidative

metabolism, whereas type II fibers have lower mitochondrial density and use glycolysis as the primary source of ATP.³ SkM biopsies from patients with HF consistently showed a lower percentage of type I oxidative fibers and a higher percentage of type II glycolytic fibers than control subjects.^{3,11,42,64,75,76} Notably, the switch to type II fibers in HF is the opposite of that observed in deconditioning, suggesting the SkM abnormalities in HF are not merely

because of inactivity.⁷⁷ The reduction in the percentage of oxidative type I fibers in HF SkM is associated with an earlier shift toward glycolytic metabolism and the accumulation of glycolytic end products.⁷⁸⁻⁸⁰ Consistent with this, reduced type I muscle fibers in patients with HF independently predicted reduced VO_2 -peak.^{11,81}

Collectively, HF is characterized by significant structural SkM abnormalities that correlate with, and likely contribute to, exercise intolerance, including reduced muscle mass, increased protein degradation, a shift from oxidative to glycolytic fiber type, reduced capillary density, and increased IMAT.

SKM METABOLIC REMODELING IN HF

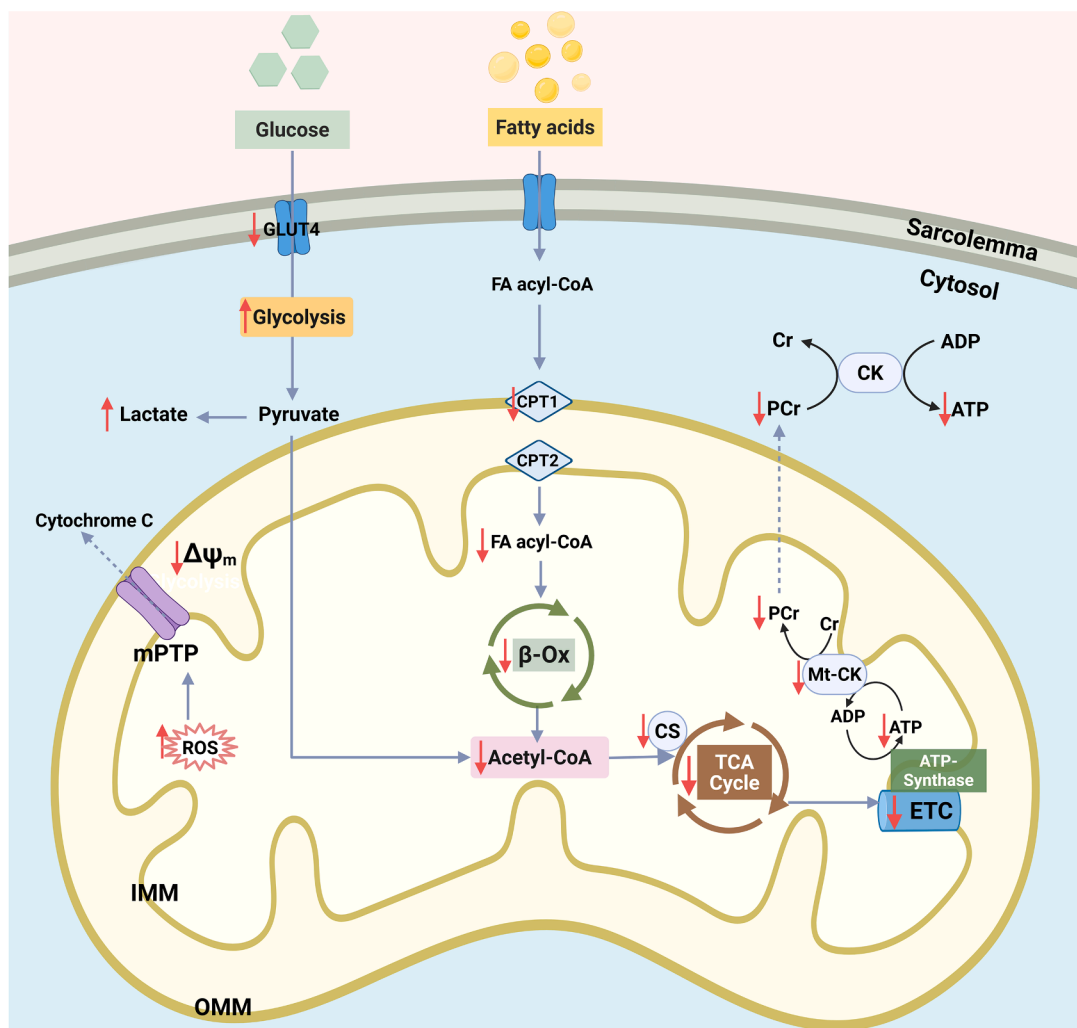
The metabolic demand of SkM can increase by 100-fold during maximal exercise; therefore, optimal mitochondrial content and function are of paramount importance for exercise performance.⁸² Studies in healthy individuals have shown that SkM mitochondrial content (quantity) and function (quality) are directly related to exercise capacity.⁸³ In HF, SkM mitochondrial content, as measured by volume density and cristae surface density, is significantly reduced and strongly correlates with a reduced VO_2 -peak.⁴² Similarly, other measures of mitochondrial content, including the expressions of porin, an abundant outer mitochondrial membrane protein, and mitofusin 2, a mitochondrial fusion protein located on the outer mitochondrial membrane, were found to be significantly reduced in SkM biopsies from patients with HF.¹² Expression of porin and mitofusin 2 correlated with VO_2 -peak and 6MWD, indicating that reduced SkM mitochondrial content and altered mitochondrial dynamics, that is an imbalance between fusion and fission, contribute to exercise intolerance in HF.¹² Consistent with this, impaired mitochondrial dynamics reflected by reduced expression of optic atrophy 1 (OPA1), a mediator of mitochondrial fusion located on the inner mitochondrial membrane, has been reported in animal models of HF and in SkM biopsies from female patients with HFpEF.^{57,66} In a healthy SkM, mitochondrial fission is followed by activation of mitophagy, which in turn accelerates the clearance of defective mitochondria, preserving a healthy mitochondrial pool.⁸⁴ The E3 ligase Parkin is a key mediator of mitophagy.⁸⁴ In patients with chronic obstructive pulmonary disease, impaired Parkin-mediated mitophagy was found to contribute to SkM atrophy.⁸⁵ Similarly, reduced Parkin-mediated mitophagy was reported in SkM in rodent models of aging.⁸⁴ However, the contribution of dysregulated

mitophagy to HF-associated SkM dysfunction is unknown. In addition to these dynamics between fusion, fission, and mitophagy, mitochondrial content is also maintained by mitochondrial biogenesis, a regenerative process regulated by peroxisome proliferator-activated receptor γ coactivator-1 α (PGC-1 α). Reduced PGC-1 α expression was recently reported in SkM biopsy from female patients with HFpEF.⁵⁷

In parallel with the reduced mitochondrial content, HF SkM is characterized by impaired oxidative capacity, associated with marked reductions in the levels and activity of several mitochondrial enzymes involved in glucose and fatty acid (FA) oxidation and oxidative phosphorylation (OXPHOS), including but not limited to, carnitine palmitoyltransferase 1, the rate-limiting enzyme for FA oxidation; citrate synthase, the rate-limiting enzyme of the tricarboxylic acid cycle; and complex I and II, key components of the electron transport chain (ETC).^{12,13,43,64,79,86-90} Direct measurements of mitochondrial function using respirometric analysis of SkM biopsy specimens from patients with HF and animal models have confirmed markedly reduced oxidative phosphorylation capacity.^{13,41} Notably, in SkM biopsies from patients with HF, the abundance and activity of the aforementioned enzymes, and the resulting reduced mitochondrial oxidative capacity, have been shown to independently correlate with measures of exercise capacity, including VO_2 -peak, VO_2 -LT, and 6MWD.^{13,40,42,43,90-92}

The reduced oxidative capacity in HF SkM leads to decreased glucose and FA oxidation, increased reliance on glycolysis, and an early lactate accumulation during exercise (Figure 2).^{13,64,78,79,93,94} In line with the impaired SkM glucose metabolism, insulin resistance is a common feature of chronic HF and has been found to correlate with SkM strength and exercise capacity.^{95,96} Oxidative stress and inflammation are implicated in SkM insulin resistance in HF.⁹⁷ Insulin resistance contributes to the reduced SkM glucose uptake and oxidation observed in HF via impaired insulin-stimulated membrane translocation of glucose transporter type 4 (GLUT4), a key facilitator of insulin-mediated glucose uptake.^{95,98,99} In addition to impaired membrane translocation, the total level of SkM GLUT4 is also reduced in HF.^{98,100}

Concomitant with reduced oxidative metabolism and OXPHOS, reduced activity of mitochondrial creatine kinase, an enzyme that transfers the high-energy phosphate PCr from mitochondria to the cytosol, is also observed in HF SkM.^{101,102} A reduced rate of PCr transfer to the cytosol accelerates exercise-induced PCr depletion, contributing to an earlier switch to glycolysis and exercise intolerance.

FIGURE 2 Adverse Skeletal Muscle Metabolic Remodeling That Occurs in Heart Failure

Skeletal muscle metabolic abnormalities in heart failure include reduced glucose and fatty acid oxidation associated with reduced expressions of the rate-limiting enzymes GLUT4 and CPT1. Enzymes involved in the Krebs cycle, such as citrate synthase (CS), as well as the electron transport chain (ETC), are reduced, thereby decreasing OXPHOS and ATP synthesis. In addition, decreased mitochondrial CK activity limits the transfer of phosphocreatine into the cytosol. Combined with decreased ATP generation, this results in significant ATP depletion, prompting a transition to anaerobic glycolysis. Other consequences of reduced OXPHOS include increased mitochondrial ROS, opening of the mPTP (mitochondrial permeability transition pore), reduced membrane potential ($\Delta\psi_m$), and release of cytochrome c, which can promote apoptosis. β -ox = β -oxidation; CK = creatine kinase; CPT = carnitine palmitoyltransferase; Cr = creatine; FA = fatty acid; GLUT4 = glucose transporter type 4; Mt-CK = mitochondrial phosphocreatine; OXPHOS = oxidative phosphorylation; TCA = tricarboxylic acid; other abbreviations as in Figure 1.

Indeed, Weiss et al,⁴⁰ using magnetic resonance spectroscopy, reported markedly reduced SkM energetic profile, as evidenced by a greater rate of PCr depletion during exercise, in both patients with HFrEF and patients with HFpEF. More recently, the same group reported that exercise intolerance in older patients with HFpEF is primarily due to rapid exercise-induced energetic decline and not due to

SkM quantity, suggesting impaired energy metabolism is the primary mechanism of SkM dysfunction and exercise intolerance in this group of patients.¹⁰³

In summary, there is compelling evidence that the SkM in HF exhibits reduced mitochondrial content and function, as well as impaired energy metabolism, each of which significantly contributes to exercise intolerance.

MECHANISMS UPSTREAM OF SKM DYSFUNCTION IN HF

NEUROHORMONAL ACTIVATION. In addition to its well-established role in the adverse cardiac remodeling and decompensated HF, activation of the renin-angiotensin-aldosterone system (RAAS) has also been shown to have a pathological role in SkM dysfunction in HF.¹⁰⁴ In SkM from animal HF models, angiotensin II (ang-II) has been shown to promote oxidative stress, induce mitochondrial dysfunction, trigger proteolysis by activating the UPS, enhance proinflammatory signaling pathways, and inhibit SkM regeneration.^{63,105-107} Ang-II-mediated signaling upregulates nicotinamide adenine dinucleotide phosphate oxidase-2, a major source of reactive oxygen species (ROS) in SkM.¹⁰⁸ Ang-II-induced increased ROS leads to mitochondrial dysfunction and activation of NOD-receptor family pyrin domain-containing protein 3 inflammasome.¹⁰⁶ Activated NOD-receptor family pyrin domain-containing protein 3 inflammasome stimulates protein degradation while inhibiting protein synthesis, thereby contributing to muscle atrophy. Thus, excessive RAAS activation is an important pathophysiologic mechanism for SkM dysfunction and exercise intolerance in HF.

OXIDATIVE STRESS. Excessive SkM ROS production, associated with altered mitochondrial structure and function, reduced muscle mass, and impaired exercise capacity, has been reported in both patients with HF and animal models.^{97,105,109} The main ROS generators in SkM during HF include mitochondrial ETC, nicotinamide adenine dinucleotide phosphate oxidase-2, and xanthine oxidase.^{60,97,105} Excessive ROS can cause direct mitochondrial damage, impair mitochondrial oxidative capacity, and induce further production of mitochondrial ROS.¹¹⁰ Excessive ROS in SkM can also inhibit anabolic pathways for protein synthesis, trigger UPS-mediated proteolysis by upregulating E3 ligases and other catabolic mediators, and promote proinflammatory cytokine expression and activity.⁶⁰ Furthermore, ROS can promote Ca²⁺ leak from the sarcoplasmic reticulum, thereby triggering cytochrome c release via the mitochondrial transition pore and activating apoptotic pathways.¹¹⁰ In HF, ROS accumulation is also promoted by reduced levels and activity of SkM antioxidants, including superoxide dismutase 2 and glutathione.¹¹¹ In animal models of HF, decreased expression of sirtuin-3, the major mitochondrial deacetylase, was associated with increased acetylation and, consequently, decreased activity of

superoxide dismutase 2, thereby contributing to oxidative stress and mitochondrial dysfunction.¹¹²⁻¹¹⁴

In addition to mitochondrial damage and triggering UPS-mediated proteolysis, excessive ROS can induce the formation of reactive lipid peroxidation products, such as acrolein, 4-hydroxynonenal (HNE), and malondialdehyde, which in turn covalently modify proteins and DNA and activate protein degradation pathways, autophagy, and apoptosis.^{115,116} Removal of these lipid peroxidation products occurs via the enzymatic activities of aldehyde dehydrogenase (ALDH2) and aldose reductase, and the scavenging actions of the nucleophile glutathione, and endogenous histidyl dipeptides such as carnosine and anserine.¹¹⁶⁻¹¹⁹ Recently, significantly elevated levels of acrolein- and 4-HNE-modified proteins were found in the gastrocnemius muscle in a mouse model of HF.¹²⁰ This was associated with decreased muscle mass and volume and reduced levels of ALDH2, carnosine, and total histidyl dipeptides in HF SkM.

Therefore, oxidative stress plays a significant pathologic role in SkM dysfunction in HF, underscoring the need to develop novel therapies that target this mechanism.

INFLAMMATION. Circulating proinflammatory cytokines such as tumor necrosis factor (TNF)- α , interleukin (IL)-6, and IL-1 β are elevated in patients with HF and contribute to SkM dysfunction and exercise intolerance.¹²¹⁻¹²⁴ The mechanisms by which inflammatory cytokines cause SkM dysfunction remain incompletely elucidated, but may include increased protein degradation via the UPS, loss of SkM cells via apoptosis, mitochondrial dysfunction via oxidative stress, insulin resistance via interference with the insulin signaling pathway, and capillary rarefaction via endothelial cell apoptosis.^{121,125} Adams et al⁵⁵ reported a correlation between increased proteasomal activity in SkM biopsies from both patients with HF_{rEF} and patients with HF_{pEF} and elevated serum IL-6 levels, suggesting that systemic inflammation contributes to SkM wasting and exercise intolerance. Serum C-reactive protein, TNF- α , and IL-6 correlated with SkM loss in patients with HF_{rEF}.^{47,126} In both patients with HF_{pEF} and patients with HF_{rEF}, serum level of the inflammatory marker growth differentiation factor (GDF)-15 was elevated and correlated with reduced exercise capacity.⁴³ In patients with HF_{rEF}, the beneficial effect of ET on alleviating exercise intolerance was associated with reductions in the local expression of inflammatory cytokines, including TNF- α , IL-6, and

IL-1 β , suggesting a pathophysiological role of local inflammation in SkM dysfunction.¹²² In line with this, excessive nitric oxide production via upregulation of inducible nitric oxide synthase in response to local inflammatory cytokines has been linked with SkM loss and impaired mitochondrial OXPHOS in animal models and humans with HF.¹²⁷⁻¹²⁹

SkM dysfunction in HF has unique features and pathological mechanisms discussed previously; however, it is important to note that HF also shares SkM abnormalities and mechanisms with several other chronic diseases and conditions, including obesity, diabetes, chronic kidney disease, chronic obstructive pulmonary disease, physical inactivity, nutritional deficiencies, and adverse medication effects. The high prevalence of comorbidities in patients with HF suggests that SkM dysfunction in HF is likely a multiple-hit disorder caused by insults associated with HF, including inflammation, oxidative stress, and excessive neurohormonal activation, as well as those associated with comorbid conditions.

OBESITY, ADIPOSITY, AND SKM DYSFUNCTION IN HF

Obesity, along with central and visceral adiposity, is a well-recognized risk factor for HFpEF.⁶⁷ Patients with obesity and HFpEF (obesity-HFpEF) have reduced bodyweight-indexed VO₂-peak compared with their nonobese counterparts due to both central and peripheral limitations that are exacerbated by obesity and regional adiposity.^{67,130,131} Obesity-associated IMAT, chronic low-grade inflammation, insulin resistance, and endothelial and microvascular dysfunction contribute to SkM dysfunction and exercise intolerance in patients with obesity-HFpEF.¹³² This, in turn, is likely because of the adipose-SkM axis, in which dysregulated adipokine secretion from chronically inflamed AT creates a proinflammatory SkM.¹³³

AGING, FRAILTY, AND SKM DYSFUNCTION IN HF

The incidence of cardiovascular (CV) diseases, including HF, increases significantly with age, coinciding with increased risk of sarcopenia, the age-related loss of SkM mass and strength, and frailty, a state of increased vulnerability to adverse health outcomes due to age-related functional decline and loss of reserve capacity.¹³⁴ On the other hand, sarcopenia and frailty are highly prevalent in patients with HF, exceeding age-adjusted estimates, and are associated with adverse clinical outcomes and poor survival in HF.^{47,134-137} The strong interrelationship among aging, frailty, sarcopenia, and HF may

reflect shared pathophysiological mechanisms underlying these conditions, including systemic inflammation, dysregulated metabolic, hormonal, and neurohormonal signaling, mitochondrial dysfunction, oxidative stress, disrupted proteostasis, and endothelial dysfunction.¹³⁸ In addition, capillary rarefaction and decline in the regenerative potential of SkM due to satellite cell exhaustion are also features shared by both aging and HF, perhaps contributing to the anabolic resistance observed in both conditions.¹³⁸

It is therefore challenging to tease out the distinct contribution of HF and aging on SkM dysfunction. For instance, it is reported that SkM of sedentary but not active older individuals have reduced OPA1, the key mitochondrial fusion protein discussed previously.¹³⁹ In adult mice, SkM-specific OPA1 deletion led to loss of SkM mass, precocious senescence, and premature death.¹³⁹ The mechanism involved increased endoplasmic reticulum stress followed by the unfolded protein response, and the induction of a catabolic program of muscle loss and systemic aging. Interestingly, pharmacological inhibition of endoplasmic reticulum stress prevented muscle atrophy and premature death. Considering the reduced OPA1 expression reported in patients with HF,⁵⁷ this exemplifies a bidirectional relationship in which aging and physical inactivity worsen HF-associated SkM dysfunction, and HF accelerates the aging phenotype. Importantly, it also suggests that therapeutic approaches targeting shared pathophysiological mechanisms between HF and aging-associated SkM dysfunction may have superior potential to improve functional capacity in patients with frailty and sarcopenia associated with these conditions.

SEX DIFFERENCES IN SKM DYSFUNCTION IN HF

Women typically have lower VO₂-peak than men because of lower cardiac output and peripheral O₂ utilization, which stem from fundamental physiologic differences, including a smaller LV cavity size, lower hemoglobin, and less SkM mass.¹⁴⁰ Similarly, compared with men, women with HF have lower VO₂-peak than men because of reduced cardiac reserve and worse peripheral impairments in O₂ extraction.¹⁴⁰⁻¹⁴³ This may explain why women with HF experience a higher burden of symptoms and worse QoL than men.¹⁴⁴ The precise mechanisms of these sex-specific differences are incompletely understood. Postmenopausal changes in body composition, including expansion of visceral adipose tissue,^{145,146}

and loss of estrogen's beneficial effects on the vasculature and mitochondrial function, could be contributing factors.¹⁴⁷⁻¹⁴⁹

Sex-specific divergence in SkM mitochondrial abnormalities has recently been reported. The expressions of OPA1 and PGC-1 α , a key regulator of mitochondrial biogenesis and oxidative capacity, were reduced in SkM biopsies only from female patients with HFrEF.⁵⁷ On the other hand, SkM biopsies from male patients with HFrEF showed lower complex I-mediated OXPHOS than those of female patients. Recently, transcriptomic analysis of SkM biopsies from patients with HFrEF showed that women, compared with men, had more type I fibers and higher SkM capillarity.¹⁵⁰

Further research is required to elucidate the mechanisms underlying the divergent sex-dependent SkM phenotypes in HF to gain insights that will inform the development of effective therapies.

THE SKM AS A THERAPEUTIC TARGET IN HF

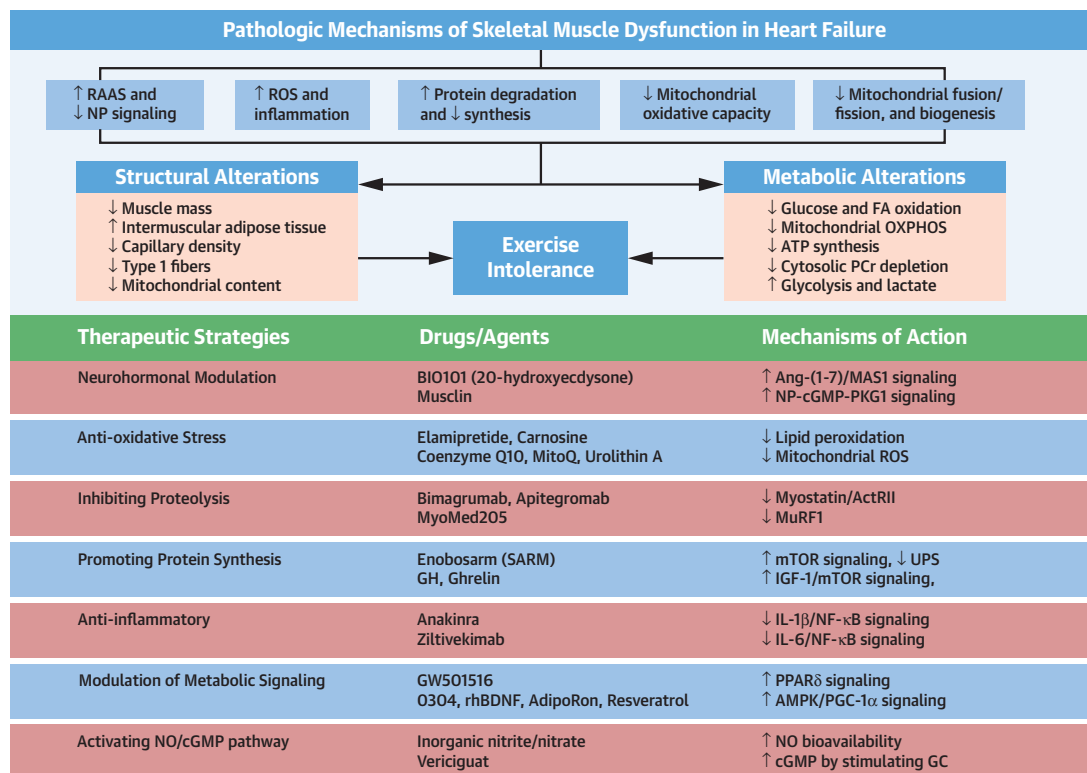
Although guideline-directed medical therapy improves outcomes in HF, consistent improvements in exercise capacity or VO₂-peak have not been demonstrated across drug classes.¹⁵¹ Most therapies have neutral or variable effects on VO₂-peak, with modest improvements observed with sodium-glucose cotransporter 2 (SGLT2) inhibitors.^{7,43,152-155} This underscores the need for novel therapies that enhance the QoL of patients with HF by improving exercise capacity. In contrast to pharmacologic therapies, ET has consistently been shown to improve exercise capacity in patients with HF by inducing favorable SkM structural and metabolic adaptations,¹⁹ suggesting that effective treatment of exercise intolerance in HF will need to use strategies that mitigate SkM dysfunction. Compared with the myocardium, SkM has a superior ability for renewal and repair,^{19,41} supporting the notion that targeting SkM could yield a rapid and meaningful improvement in exercise and functional capacity. In the following, we provide a discussion of the role of exercise and other potential therapeutic approaches that target SkM dysfunction in HF (**Central Illustration**).

EXERCISE. Exercise training is associated with improved exercise capacity, QoL, and clinical outcomes in both HFrEF and HFpEF.^{6,19,33,34,36,46,156-160} The improvement in VO₂-peak after ET in patients with HF has been shown to be primarily driven by improved SkM O₂ extraction and utilization, rather than by changes in central hemodynamic parameters, suggesting that exercise induces favorable metabolic

and morphological SkM adaptations that mitigate SkM dysfunction in HF.^{35,36,75,161} The ET-associated increase in SkM O₂ consumption could be due to improved O₂ delivery from increased capillary density and endothelial function, or to improved O₂ utilization related to increased mitochondrial content and function.^{75,161-163} Mechanistically, evidence from experimental and clinical studies indicates that exercise ameliorates SkM dysfunction via its effects on several cellular processes, including antioxidative stress,^{164,165} anti-inflammatory,¹²² restoration of the balance between catabolic and anabolic pathways of protein turnover,^{56,165,166} enhancing mitochondrial oxidative capacity and promoting mitochondrial biogenesis,^{161,167-170} and increasing capillary density by inducing SkM angiogenesis.¹⁶³ In addition, experimental and clinical evidence show that exercise increases the number and activity of satellite cells, thereby enhancing SkM regenerative capacity.^{61,171-175}

Based on evidence from large randomized clinical trials (RCTs), such as HF-ACTION (Heart Failure: A Controlled Trial Investigating Outcomes of Exercise Training) and several others supporting significant improvements in exercise capacity with ET programs in patients with HF,^{9,160} supervised ET is now a class I recommendation in the management guidelines for HF.³³ Despite this, the lack of universal access to supervised training programs and the challenges of long-term adherence remain significant limitations.^{33,159} Nonetheless, the salutary effects of exercise on SkM and mitochondrial function discussed previously represent promising targets for novel therapies.

CALORIC RESTRICTION. Caloric restriction (CR), that is, a 30% to 40% reduction in caloric intake, is a well-established strategy for weight loss and has been shown to have favorable effects on systemic inflammation, oxidative stress, metabolic parameters, and SkM homeostasis (**Table 1**).^{176,177} In patients with obesity-HFpEF, 5 months of CR with or without aerobic ET significantly improved VO₂-peak¹⁵⁷ and the combination of CR and aerobic ET produced a robust increase in VO₂-peak, suggesting synergistic effects.¹⁵⁷ Consistent with previous studies linking exercise intolerance in patients with obesity-HFpEF to whole-body, visceral, and SkM adiposity,^{67,68,178} the increase in VO₂-peak with CR correlated with increased percent lean body mass and thigh muscle-to-IMAT ratio, indicating that reduced adiposity contributes to improved exercise capacity.¹⁵⁷ Despite improved exercise capacity, however, CR, with or without aerobic ET, was associated

CENTRAL ILLUSTRATION Exercise Intolerance in Heart Failure Due to Skeletal Muscle Dysfunction

Challa AA, et al. JACC Heart Fail. 2026;■(■):103353.

Exercise intolerance in heart failure is associated with skeletal muscle dysfunction characterized by significant structural and metabolic alterations. Structural skeletal muscle abnormalities in heart failure include reduced muscle mass, a shift in fiber-type composition, increased intermuscular adipose tissue, capillary rarefaction, and reduced mitochondrial content. Metabolic skeletal muscle alterations in heart failure include reduced glucose and fatty acid (FA) oxidation, increased glycolysis and lactate production, impaired mitochondrial oxidative phosphorylation (OXPHOS), reduced ATP synthesis, and increased rate of phosphocreatine (PCr) depletion. The pathologic mechanisms of skeletal muscle dysfunction in heart failure involve excessive neurohormonal activation, oxidative stress, and inflammation, accompanied by enhanced proteolysis and mitochondrial dysfunction. Aside from exercise, there are currently no therapies with proven efficacy in improving exercise capacity in patients with heart failure. Emerging therapies targeting the key mechanisms of skeletal muscle dysfunction have shown promising results in experimental studies. ActRII = activin type II receptor; AdipoRon = adiponectin receptor agonist; AMPK = AMP-activated protein kinase; Ang-(1-7) = angiotensin-(1-7); cGMP = cyclic guanosine monophosphate; GH = growth hormone; IGF = insulin-like growth factor; IL = interleukin; MAS1 = angiotensin-(1-7) receptor; mTOR = mechanistic target of rapamycin; MuRF1 = protein muscle ring finger 1; NF-κB = nuclear factor kappa-light-chain-enhancer of activated B cells; NO = nitric oxide; NP = natriuretic peptide; O304 = pan-AMPK activator; PCr = phosphocreatine; PGC-1α = peroxisome proliferator-activated receptor-γ coactivator 1-alpha; PKG = protein kinase G; PPARδ = peroxisome proliferator-activated receptor-delta; RAAS = renin-angiotensin-aldosterone system; rhBDNF = recombinant human brain-derived neurotrophic factor; ROS = reactive oxygen species; SARM = selective androgen receptor modulator; sGC = soluble guanylate cyclase; UPS = ubiquitin-proteasome system.

with a loss of SkM mass.^{157,179} Interestingly, adding resistance ET to CR and aerobic ET increased leg strength and muscle quality without attenuating SkM loss.¹⁷⁹ At long-term follow-up, patients with HFpEF who participated in CR regained substantial weight, predominantly adipose tissue, and showed significantly reduced VO₂-peak compared with their values

at the end of the 5-month CR protocol,¹⁸⁰ highlighting the need for long-term adherence when using CR as a therapeutic strategy. In addition, whether CR poses a risk of sarcopenia in non-obese older patients with HF, similar to that reported in older healthy individuals,¹⁸¹ needs further investigation.

DIETARY INTERVENTIONS. The DASH (Dietary Approaches to Stop Hypertension) diet reduced systemic oxidative stress and improved exercise capacity in patients with HFpEF in a small pilot study.¹⁸² In experimental studies, the Mediterranean diet and diets rich in unsaturated fatty acids (UFAs) have been shown to exert SkM-protective effects by mitigating inflammation, oxidative stress, and atrophic signaling (Table 1).¹⁸³⁻¹⁸⁵ However, clinical evidence for their effect on exercise capacity in HF is limited. Although dietary n-3 FA supplementation for a year was not associated with improved exercise capacity in patients with HFpEF,¹⁸⁶ increased plasma UFA levels following 12 weeks of a high-UFA diet correlated with improved VO₂-peak in patients with obesity-HFpEF.¹⁸⁷ Experimental evidence supports the protective role of amino acid supplementation against HF-associated SkM dysfunction (Table 1).¹⁸⁸ In older patients with HFrEF with evidence of sarcopenia, a few small clinical trials reported that supplementation with essential amino acids increased lean muscle mass, but with mixed results on improvement in muscle strength or exercise performance.^{189,190}

As discussed previously, carnosine, a dipeptide endogenously synthesized from β -alanine and histidine, is a free-radical scavenger and a potent antioxidant abundantly present in SkM.^{118,119,191} SkM carnosine content declines with aging and disease states and can be readily increased by dietary intake or exogenous supplementation. In older adults, β -alanine supplementation was associated with increased SkM carnosine content and improvements in exercise tolerance.^{120,192} Importantly, in an RCT involving stable patients with HFrEF, carnosine supplementation for 6 months significantly improved VO₂-peak, VO₂-LT, 6MWD, and QoL score.¹⁹³ This promising result has not been followed up with large-scale trials; in addition, carnosine supplementation has not been tested in HFpEF. The beneficial effects of carnosine on exercise tolerance in older individuals and in patients with HFrEF suggest that aldehyde detoxification in SkM may be critical in these populations. To this effect, Chaudhari et al¹²⁰ investigated whether toxic aldehydes accumulate in the SkM during HF using a mouse model of pressure overload-induced HF. The authors observed that acrolein and 4-HNE adduct levels were significantly elevated in SkM 12 weeks after transaortic constriction, accompanied by decreased muscle weight, strength, and ALDH2 expression.¹²⁰ Importantly, carnosine levels were also decreased in the atrophic muscle. They next examined whether SkM carnosine levels decline

before the onset of overt HF. Mice subjected to transaortic constriction for 4 weeks showed reduced SkM carnosine levels even before measurable changes in cardiac function, suggesting that reduced SkM carnosine levels are not the result of hemodynamic abnormalities accompanying decompensated HF. Correcting SkM carnosine depletion early in the course of HF could prevent exercise intolerance that often follows as HF progresses. The fact that β -alanine and carnosine are over-the-counter supplements is significant, as they could represent an accessible, cost-effective solution to exercise intolerance for patients with HF.

Accumulating evidence supports the beneficial effects of therapeutic ketosis in HF; however, it remains unclear whether exogenous ketone bodies improve exercise capacity in HF.¹⁹⁴ Recently, acute administration of a ketone ester drink was reported to have no effect on VO₂-peak in patients with HFpEF, despite increasing cardiac output and reducing LV filling pressure.¹⁹⁵ Investigation of the long-term administration of exogenous ketone bodies in trials with exercise capacity as an endpoint is warranted. Similarly, the effect of a ketogenic diet on exercise capacity in HF is unknown.

Ongoing trials are assessing the effects of several dietary strategies on exercise capacity in patients with HF (Table 2).

GLP-1 RAs. In patients with obesity-HFpEF with or without diabetes, semaglutide, the GLP-1 RA, or tirzepatide, a combined agonist of the glucose-dependent insulinotropic polypeptide and GLP-1 receptors, improved symptoms, CV outcomes, and functional capacity, as measured by 6MWD.¹⁹⁶⁻¹⁹⁸ Post hoc analyses showed improvements in 6MWD with semaglutide and tirzepatide correlated with reductions in body weight, blood volume, and systemic inflammation.¹⁹⁹⁻²⁰¹ In animal models of HFpEF, GLP-1 RAs improved cardiac function and exercise capacity associated with reductions in inflammation and oxidative stress, effects not observed with CR alone, suggesting mechanisms beyond weight loss.^{202,203} In line with this, GLP-1 RAs have been reported to reduce IMAT, restore the balance between SkM protein degradation and synthesis, improve insulin sensitivity, and enhance mitochondrial content and function.²⁰⁴⁻²⁰⁸

GLP-1 RA-induced weight loss has caused concerns over the associated risk of sarcopenia in vulnerable individuals.²⁰⁹ This is primarily because of reports from RCTs indicating that a significant portion of GLP-1 RA-induced weight loss is attributable to loss of lean body mass.²¹⁰⁻²¹² In addition to SkM, however,

TABLE 1 Therapeutic Potential of Pharmacological and Nutritional Strategies to Treat SkM Dysfunction in HF: Experimental Evidence

Drug/Strategy	Description/Class	Experimental Model	Effects on SkM Structure, Metabolism, and Function	First Author
AdipoRon	A small molecule agonist of adiponectin receptor	Diet-induced obesity in AdipoR-humanized mice	↑ AMPK, ↑ PGC-1 α , ↑ expression of oxidative metabolism and OXPHOS, ↑ SOD2, ↑ insulin sensitivity, ↑ type 1 fibers, ↑ exercise capacity	Iwabu et al ²⁹³
Bimagrumab	Human mAb that inhibits ActRII receptors	Diet-induced obesity (mouse)	↓ myostatin signaling, ↓ protein degradation, ↑ muscle mass, ↑ exercise capacity	Nunn et al ²¹⁸
BIO101	20-hydroxyecdysone, an ecdysteroid	Age-related sarcopenia (mouse)	↑ Activation of Ang-(1-7)/MAS1 receptor, ↑ mitochondrial fusion, ↑ muscle mass, ↑ exercise capacity	Serova et al ²³⁰
Butyrate	SCFA	Diabetic nephropathy (db/db) (mouse)	↑ GPR43 mediated Akt/mTOR signaling, ↑ muscle mass, ↓ ROS, ↑ mitochondrial OXPHOS, ↑ exercise capacity	Tang et al ²⁹⁸
Caloric restriction	40% decrease in total calories and daily resistance exercise	Obese ZSF1 HFpEF model (rat)	↑ mitochondrial function and coupling efficiency, ↑ mitochondrial OXPHOS, ↑ myofiber size, ↑ myonuclear accretion, ↑ myocyte regeneration	Espino-Gonzalez et al ⁶¹
Carnosine	A dipeptide made of β -alanine and histidine	Hypobaric hypoxia-induced myopathy (rat)	↓ lipid peroxidation products, ↓ inflammation, ↓ apoptosis ↓ protein degradation	Agrawal et al ²⁹⁹
Elamipretide	Binds and stabilizes cardiolipin at IMM	Post-MI HF (dog)	↓ cardiolipin peroxidation, ↓ ROS, ↓ inflammation ↑ type 1 muscle fiber, ↑ mitochondrial OXPHOS	Sabbah et al ²⁷³
Extra virgin olive oil	An element of the Mediterranean diet	Diet-induced obesity (rat)	↑ mitochondrial fusion/↓ fission, ↑ antioxidant defense, ↓ protein degradation, ↑ mitochondrial content, ↑ OXPHOS enzymes	Yeo et al ¹⁸³
Febuxostat	An XO inhibitor	Post-MI HF (mouse)	↓ ROS, ↑ mTOR signaling, ↑ protein synthesis, ↑ type 1 fibers, ↑ mitochondrial OXPHOS, ↑ exercise capacity	Nambu et al ⁶⁰
Ghrelin (acylated)	A gastric hormone with orexigenic effects	Post-MI HF (rat)	↑ insulin signaling, ↓ inflammation signaling, ↑ mitochondrial oxidative enzymes, ↑ muscle mass	Barazzoni et al ²⁵⁴
GW501516	PPAR δ agonist	Post-MI HF (mouse)	↑ CPT1 expression, ↑ FA oxidation, ↑ ATP level, ↑ exercise capacity	Zizola et al ³⁰⁰
Honokiol	A natural compound and activator of SIRT3	Post-MI HF (mouse)	↑ SIRT3 level, ↓ SOD2 acetylation, ↓ ROS, ↑ mitochondrial OXPHOS, ↑ exercise capacity	Masunaga et al ¹¹¹
Mito-TEMPO	Mitochondria-targeted antioxidant	Post-MI HF (rat)	↓ diaphragm mitochondrial H ₂ O ₂ ↑ diaphragmatic muscle strength	Laitano et al ³⁰¹
Leucine supplement	A diet containing 3% leucine	Obese ZSF1 HFpEF (rat)	↑ mitochondrial content and oxidative capacity ↑ exercise capacity	Alves et al ¹⁸⁸
MyoMed-205	Small molecule inhibitor of MuRF1	Obese ZSF1 HFpEF model (rat)	↓ muscle protein ubiquitination, ↑ complex I and II protein level, ↑ muscle mass, ↑ muscle strength	Adams et al ³⁰²
Musclin	A myokine that binds NPRC to limit NP degradation	SAUNA HFpEF model (mouse)	↑ NP/cGMP/PKG1 signaling, ↑ PGC-1 α expression, ↑ type 1 fiber, ↑ complex II activity, ↑ capillary density (soleus), ↑ exercise capacity	Saw et al ²³⁴
n-3 PUFA-enriched diet	EPA and DHA (27% of total fat intake)	Post-MI HF (mouse)	↓ ROS, ↓ inflammation, ↑ insulin signaling, ↑ mitochondrial OXPHOS, ↑ muscle mass	Gortan et al ⁶⁶
O304	Pan-AMPK activator	Age-associated sarcopenia (mice)	↑ insulin sensitivity, ↑ SkM glucose uptake, ↑ exercise capacity	Ericsson et al ²⁸⁶
Resveratrol	A natural polyphenol activator of AMPK	Pressure overload-induced HF (mouse)	↑ insulin sensitivity, ↑ mitochondrial OXPHOS, ↑ exercise capacity	Sung et al ²⁸⁸
rhBDNF	A neurotrophic growth factor	Post-MI HF (mouse)	↑ AMPK α and PGC-1 α , ↑ mitochondrial biogenesis, ↑ FA oxidation, ↑ mitochondrial OXPHOS, ↑ VO ₂ -peak	Matsumoto et al ²⁹¹
Trimetazidine	Inhibits FA β -oxidation	Diet-induced obesity (mouse)	↑ glucose oxidation and ↑ mitochondrial OXPHOS, ↑ mitochondrial fusion and biogenesis, ↓ atrogin1, ↓ MuRF1, ↑ muscle mass, ↑ exercise capacity	Zhang et al ³⁰³
Vericiguat	Stimulates sGC to activate cGMP and PKG1	Diet-induced obesity in apoE ^{-/-} (mouse)	↓ mitochondrial ROS, ↑ antioxidant defense, ↓ IL-6, ↓ TNF- α , ↑ mitochondrial content, ↑ type 1 muscle fiber, ↑ muscle mass, ↑ exercise capacity	Li et al ²⁴³

ActRII = Activin type II receptor; AdipoR = adiponectin receptor; Akt = protein kinase B; AMPK = AMP-activated protein kinase; Ang-(1-7) = angiotensin-(1-7); apoE = apolipoprotein E; ATP = adenosine triphosphate; cGMP = cyclic guanosine monophosphate; CPT1 = carnitine palmitoyltransferase 1; DHA = docosahexaenoic acid; EPA = eicosapentaenoic acid; FA = fatty acid; GPR43 = G-protein coupled receptor 43; HF = heart failure; HFpEF = heart failure with preserved ejection fraction; IMM = inner mitochondrial membrane; IL = interleukin; mAb = monoclonal antibody; MAS1 = angiotensin-(1-7) receptor; MI = myocardial infarction; mTOR = mechanistic target of rapamycin; MuRF1 = muscle ring finger protein-1; NP = natriuretic peptide; NPRC = natriuretic peptide receptor C; OXPHOS = oxidative phosphorylation; PGC-1 α = peroxisome proliferator-activated receptor- γ coactivator 1- α ; PKG1 = protein kinase G1; PPAR δ = peroxisome proliferator-activated receptor- δ ; PUFA = polyunsaturated fatty acid; rhBDNF = recombinant human brain-derived neurotrophic factor; ROS = reactive oxygen species; SAUNA = salty drinking water/unilateral nephrectomy/aldosterone; SCFA = short-chain fatty acid; sGC = soluble guanylate cyclase; SIRT3 = sirtuin 3; SkM = skeletal muscle; SOD2 = superoxide dismutase 2; TNF = tumor necrosis factor; VO₂-peak = peak oxygen consumption; XO = xanthine oxidase; ZSF1 = Zucker fatty/spontaneously hypertensive heart failure F1 hybrid.

lean body mass includes internal organs, bones, and other tissues.^{209,213} Lean muscle mass loss may not always mean loss of strength or functional capacity. For instance, in the SEMALEAN (Impact of

Semaglutide on fat mass, lean mass and muscle function in patients with obesity: The SEMALEAN study) prospective cohort, semaglutide significantly improved handgrip strength at 12 months despite

TABLE 2 Ongoing Clinical Trials Investigating Pharmacologic and Nutritional Strategies With a Potential for Treating SkM Dysfunction in HF

Intervention	Clinical Trial	Phase	Study Population	Duration	SkM-Related Outcomes	Remark
Bimagrumab	A Study to Investigate Weight Management With Bimagrumab (LY3985863) and Tirzepatide (LY3298176), Alone or in Combination, in Adults With Obesity or Overweight (NCT06643728)	2	N = 252; adults with obesity or overweight who are taking tirzepatide for weight loss	48 wk	Lean body mass, measured by DEXA	
	A Study of Bimagrumab (LY3985863) and Tirzepatide (LY3298176), Alone or in Combination, in Participants With Obesity or Overweight With Type 2 Diabetes (NCT06901349)	2	Patients with type 2 diabetes and obesity	36 wk	Lean body mass, measured by DEXA	Terminated due to "strategic business reasons"
BIO101	OBA (NCT07411378)	2	N = 164; adults with obesity who are taking semaglutide for weight loss	21 wk	Muscle strength, 6MWD, and lean body mass, measured by DEXA	BIO101 is 20-hydroxyecdysone
Colchicine	COLEHEART-PRESERVED (NCT06081049)	2	N = 155; patients with HFpEF	6 mo	6MWD	
	CHIPS (NCT06604611)	4	N = 200; patients with HFpEF and hsCRP > 2 mg/L	12 wk	6MWD	
Enobosarm	PLATEAU (NCT07446998)	2b	N = 200; patients with obesity (BMI ≥35 kg/m ²) and age ≥65 y, who are taking semaglutide for weight loss	68 wk	Lean body mass, measured by DEXA	
MitoQ	Mito-Frail (NCT06027554)	2	N = 60; older adults and frail older adults with physical and/or cognitive dysfunction	12 wk	Gait speed, endothelial function via brachial artery FMD	Secondary endpoints include measures of systemic oxidative stress
Potassium nitrate	iNIX-HF (NCT05562167)	2	N = 75; patients with HFref	6 wk	VO ₂ -peak via CPET and maximal muscle power (quadriceps)	Secondary endpoints include exercise time
S-pindolol benzoate	PROACT 1 (NCT07101939)	2	N = 120; adults with obesity who are taking semaglutide for weight loss	20 wk	Lean body mass, measured by DEXA, muscle strength, muscle power, 6MWD	
	PROACT 2 (NCT07101939)	2	N = 120; adults who discontinued semaglutide (after 20 wk of therapy in the PROACT 1 trial)	20 wk	Lean body mass, measured by DEXA, muscle strength, muscle power, 6MWD	
Semaglutide +/- exercise training	Effect of GLP-1 Receptor Agonist Therapy With and Without Exercise Training on Muscle Mass and Physical Function in People With Obesity (NCT07091500)	NA	N = 40; adults with obesity and decreased physical function	52 wk	Physical performance test, lean body mass by DEXA	Secondary outcomes include MRI scans for SkM volume and muscle strength tests
Taldefgrobep alfa	Biohaven Therapeutics (NCT07281495)	2b	N = 15; adults with obesity	24 wk	Lean body mass, measured by DEXA	
Trevogrumab + garetosumab	COURAGE (NCT06299098)	2	N = 1,005; adults with obesity who are taking semaglutide for weight loss	26 wk	Lean body mass, measured by DEXA	

Continued on the next page

an initial loss of lean mass.²¹⁴ Unfortunately, most published clinical trials have not directly assessed SkM mass or strength.²¹⁵ And animal studies have reported conflicting results regarding changes in SkM mass and strength with GLP-1 RA therapy.²¹⁶⁻²¹⁹ Therefore, the extent of GLP-1 RA-induced SkM mass loss remains unsettled and warrants further investigation.²⁰⁹ Importantly, in older patients with HFpEF, whether the

forementioned beneficial SkM effects of GLP-1 RAs outweigh the potential adverse effect of SkM mass loss remains to be elucidated.

SGLT2 INHIBITORS. Experimental studies have shown that SGLT2 inhibitors ameliorate SkM dysfunction in HF by enhancing mitochondrial oxidative metabolism, preventing muscle atrophy, and reducing inflammation and oxidative stress.²²⁰⁻²²² Recently, in a study of SkM biopsies from patients

TABLE 2 Continued

Intervention	Clinical Trial	Phase	Study Population	Duration	SkM-Related Outcomes	Remark
Ziltivekimab	ATHENA (NCT06200207)	3	N = 680; patients with HFpEF or HFmrEF and serum hsCRP $\geq$ 2 mg/L	12 mo	6MWD	
DASH diet + HIIT	The Effect of a High-Fiber Diet and High-Intensity Interval Exercise in Patients With HFpEF (NCT05236413)	NA	N = 36; patients with HFpEF, NYHA functional class II-III symptoms	4 wk	VO ₂ -peak	
High-protein diet (ensure maximum protein)	ASTRID-HF (NCT05627440)	NA	N = 120, patients with HFpEF, NYHA functional class II-IV	6 mo	Lean mass as measured by DEXA, handgrip strength, and 6MWD	Secondary outcome measures include GDF-15, activin-A, and FSTL-3
Ketogenic diet	STAK (NCT06081543)	NA	N = 90; patents with HFpEF and diabetes or pre-diabetes, or metabolic syndrome, or obesity	26 wk	VO ₂ -peak	

6MWD = 6-minute walk distance; ASTRID-HF = A Skeletal Muscle Recovery Intervention With Dietary Protein in Heart Failure; ATHENA = A Research Study Looking Into How Ziltivekimab Works Compared to Placebo in Participants With Heart Failure and Inflammation; BMI = body mass index; CHIPS = Colchicine in Patients With Heart Failure With Preserved Ejection Fraction and Inflammation; COLHEART-PRESERVED = The Colchicine Heart Failure PRESERVED Trial; COURAGE = A Study to Test if Trevogrumab or Trevogrumab With Garetosmab When Taken With Semaglutide is Safe and How Well They Work in Adult Patients With Obesity for Weight Loss, Fat Loss, and Lean Mass Preservation; CPET = cardiopulmonary exercise test; DASH = dietary approaches to stop hypertension; DEXA = dual-energy x-ray absorptiometry; FMD = flow-mediated dilation; FSTL-3 = Follistatin-like 3; GDF = growth differentiation factor; HFmrEF = heart failure with mildly reduced ejection fraction; HFpEF = heart failure with preserved ejection fraction; HFrEF = heart failure with reduced ejection fraction; HIIT = high-intensity interval training; hsCRP = high sensitivity C-reactive protein; INX-HF = Inorganic Nitrate and Exercise Performance in Heart Failure trial; Mito-Frail = The Effects of MitoQ on Vasodilation, Mobility, and Cognitive Performance in Frail Older Adults; MRI = magnetic resonance imaging; NA = not applicable; OBA = A Phase 2 Clinical Development Program Evaluating BIO101 (20-hydroxyecdysone) for obesity; PLATEAU = A Phase 2 Clinical Trial Evaluating the Effect on Body Composition and Safety of Enobosarm in Patients Treated with GLP-1 Receptor Agonists for Chronic Weight Management; PROACT 1 and 2 = A Study of the Effect of (S)-Pindolol Benzoate on Lean Body Mass in Obese Patients During (PROACT 1), and Post-semaglutide Therapy (PROACT 2); SkM = skeletal muscle; STAK = Ketogenic Diet vs Mixed Diet in Patients With Heart Failure; VO₂-peak = peak O₂ consumption.

with HFrEF, Wood et al²²³ reported that treatment with SGLT2 inhibitors was associated with antiatrophic, anti-inflammatory, and prometabolic effects. However, results from clinical studies on the effect of SGLT2 inhibitors on exercise intolerance in HF have been mixed.^{153,224} Although dapagliflozin improved 6MWD in the PRESERVED-HF (Dapagliflozin in PRESERVED Ejection Fraction Heart Failure) trial,²²⁵ it showed neutral effects on 6MWD in the DETERMINE (Dapagliflozin Effect on Exercise Capacity Using a 6-Minute Walk Test in Patients With Heart Failure) trial.²²⁶ A recent meta-analysis reported modest improvements in 6MWD with SGLT2 inhibitor therapy in patients with HF regardless of ejection fraction.¹⁵²

NEUROHORMONAL MODULATORS. As discussed previously, preclinical evidence supports the role of excessive RAAS activation in HF-associated SkM dysfunction. Consistently, RAAS inhibition, with angiotensin-converting enzyme inhibitors or ang-II receptor blockers, was associated with a lower prevalence of muscle mass loss in patients with HF.²²⁷ However, clinical studies investigating the effect of RAAS inhibition on exercise capacity in HF report conflicting results. BIO101 is a pharmaceutical formulation of 20-hydroxyecdysone, a plant-derived ecdysteroid that activates the Ang-(1-7) receptor, Mas-receptor.²²⁸ Ang-(1-7), a functional antagonist to ang-II, was shown to prevent ang-II-induced SkM atrophy in mice.²²⁹ In a mouse model of aging-

induced sarcopenia, BIO101 improved SkM strength and exercise performance.²³⁰ In the phase 2 SARA-INT (Sarcopenia Age-related RArcopenia [SARA] INTerventional Clinical Trial), the higher dose regimen of BIO101 350 mg twice daily for 6 to 9 months improved 400-m gait speed in older patients with sarcopenia.²³¹ Currently, the Phase 2 OBA (Testing the Efficacy, Safety, and PK of 20E in Patients With Obesity Who Are Starting Treatment With the GLP-1 Agonist Semaglutide for Weight Loss; NCT07411378) is investigating the efficacy of BIO101 in preventing lean body mass loss in patients with obesity taking semaglutide for weight loss (Table 2).

Musclin, an exercise-responsive myokine, has been shown to enhance SkM oxidative capacity by binding to natriuretic peptide receptor C, thereby limiting natriuretic peptide (NP) degradation and potentiating NP signaling.²³² Augmented NP signaling in SkM has been shown to induce PGC-1 α via cyclic guanosine monophosphate/protein kinase G1 pathway and enhances oxidative capacity.²³³ Exogenous administration of musclin in HFpEF mice potentiated NP signaling, promoted fiber type shift from type 2 to type 1, increased mitochondrial oxidative capacity, and improved exercise capacity.²³⁴ The therapeutic potential of musclin has not yet been tested in clinical trials.

NITRIC OXIDE/CYCLIC GUANOSINE MONOPHOSPHATE MODULATORS. Acute or short-term administration

of inorganic nitrite or nitrate improved exercise capacity in patients with HFpEF and HFrEF.²³⁵⁻²³⁸ However, chronic inorganic nitrate/nitrite use in patients with HFpEF failed to improve exercise capacity or QoL in large-scale trials.^{29,30,239} An ongoing phase 2 RCT is investigating the efficacy of potassium nitrate on exercise capacity in patients with HFrEF.²⁴⁰ Vericiguat, a soluble guanylate cyclase stimulator, is Food and Drug Administration-approved for use in HFrEF based on clinical evidence that showed reductions in hospitalization and mortality in patients with high-risk HFrEF.²⁴¹ In a small prospective observational study involving 79 patients with HFrEF, vericiguat added to guideline-directed medical therapy significantly improved VO₂-peak.²⁴² Although vericiguat was shown to prevent obesity-induced exercise intolerance in a rodent model by reducing oxidative stress and boosting mitochondrial function,²⁴³ results from clinical trials investigating the effect of vericiguat on exercise capacity in patients with HFpEF have thus far been negative.²⁴⁴

ANABOLIC HORMONES AND PHARMACEUTICALS. Anabolic hormones stimulate SkM protein synthesis and promote muscle growth; hence, they have been tested as a strategy to improve SkM function in HF. Clinical trials on the effect of testosterone supplementation in patients with HF are inconclusive but suggest possible beneficial effects on exercise capacity in those with sarcopenia and hypogonadism.^{245,246} However, given concerns over potential adverse effects,²⁴⁷ large-scale trials are needed to confirm the safety and efficacy of testosterone therapy in patients with HF.

Selective androgen receptor modulators (SARMs) bind and activate androgen receptors in SkM and bone, sparing the prostate and other androgen-sensitive tissues, hence minimizing off-target effects. In SkM, SARMs stimulate protein synthesis by activating mechanistic target of rapamycin signaling and inhibiting UPS-mediated proteolysis. Enobosarm, a nonsteroidal SARM, increased lean mass and improved muscle strength in preclinical models, healthy older men, postmenopausal women, and patients with cancer cachexia.²⁴⁸⁻²⁵⁰ Recently, the phase 2b QUALITY (Evaluating Enobosarm Combined With Semaglutide for Muscle Preservation and Fat Loss) trial showed that adding enobosarm 3 mg daily for 16 weeks to semaglutide preserved 99% of lean mass in older adults with obesity, suggesting it facilitated highly selective fat mass loss. The ongoing phase 2b PLATEAU (Phase 2b PLATEAU Clinical Trial of Enobosarm and

Semaglutide Combination for High Quality Weight Loss) aims to validate the results from QUALITY with a longer 68-week treatment in older adults with class 2 obesity (body mass index ≥ 35).

Growth hormone (GH) and IGF-1 exhibited pro-anabolic effects on SkM in preclinical studies⁵⁴; however, clinical evidence of their effects on SkM dysfunction in patients with HF has been inconsistent.²⁵¹ In a recent small RCT, 1 year of GH replacement in patients with HFrEF with GH deficiency improved VO₂-peak, 6MWD, and handgrip strength.²⁵² Interestingly, the study reported a 27% prevalence of GH deficiency in patients with HFrEF and suggested that inconsistencies in GH trials in HF may be because of previous trials not being targeted to the subset of patients with GH deficiency. Ghrelin and its analogs prevent SkM loss in preclinical studies and in patients with cancer cachexia.²⁵³⁻²⁵⁵ In a small clinical trial, short-term (3 weeks) intravenous administration of ghrelin improved LV function, VO₂-peak, lean mass, and muscle strength in patients with HFrEF.²⁵⁶ Espindolol (S-pindolol benzoate) is a partial- β_2 adrenergic agonist, nonselective β antagonist, and 5-HT_{1A} receptor agonist that has been shown to increase muscle mass in animal studies and in patients with cancer cachexia.^{257,258} The phase 2a PROACT (Preserving, Restoring, and Optimising [lean mass and muscle] with ACTAs) is investigating the efficacy of S-pindolol to preserve lean mass during GLP-1 RA therapy (Table 2).

The potential of the preceding pro-anabolic agents to improve exercise capacity in patients with HF warrants confirmation in large-scale RCTs.

MYOSTATIN/ACTIVIN INHIBITORS. Bimagrimab, a human monoclonal antibody that targets ActRIIB, inhibits the activities of myostatin and activin-A and promotes SkM growth.^{218,259} In patients with obesity and diabetes, intravenous bimagrimab every 4 weeks for 48 weeks reduced body weight by 6.5%, decreased fat mass by 20.5%, and increased lean mass by 3.6%.²⁶⁰ In the BELIEVE (Phase 2b study evaluating bimagrimab alone and in combination with semaglutide in patients with overweight or obesity) trial, the addition of high-dose bimagrimab (30 mg/kg every 12 weeks) to semaglutide therapy preserved lean mass, with 92.3% of weight loss attributable to fat loss, compared with 71.1% in semaglutide monotherapy.²⁶¹ A phase 2 RCT is currently investigating the efficacy of brimagrimab in preserving lean mass during tirzepatide-induced weight loss (Table 2).

Apitegromab, a highly selective myostatin antagonist targeting the latent form of myostatin, has been

shown to improve motor function in patients with spinal muscular atrophy.²⁶² EMBRAZE (Apitegromab for lean mass preservation during tirzepatide-induced weight loss: a randomized, double-blind, placebo-controlled phase 2 trial) reported that apitegromab resulted in 1.9 kg less lean mass loss than placebo, equivalent to a 54.9% preservation of lean mass, during tirzepatide therapy for weight loss.²⁶³ In the phase 3 RESILIENT (A Randomized, Double-Blind, Placebo-Controlled Study to Evaluate the Efficacy and Safety of Taldefgrobep Alfa in Participants With Spinal Muscular Atrophy) trial, taldefgrobep alfa, a recombinant protein that neutralizes myostatin, increased lean mass but did not improve motor function in patients with spinal muscular atrophy.²⁶⁴ Trevogrumab and garetosumab are monoclonal antibodies that target myostatin and activin-A, respectively. Based on positive results in preclinical studies,²⁶⁵ the phase 2 COURAGE (A Randomized, Double-Blind Study of the Efficacy and Safety of Trevogrumab, With or Without Garetosumab, in Addition to Semaglutide in Patients With Obesity) trial is investigating the effect of combination therapy with trevogrumab and garetosumab on lean mass preservation during GLP-1 RA therapy for weight loss.

Although the preceding results show the potential of myostatin/activin inhibitors to preserve muscle mass during weight-reduction therapies, whether this translates into gains in exercise capacity, QoL, or improved clinical outcomes remains to be seen. Furthermore, the efficacy of myostatin inhibitors on HF-associated SkM dysfunction is unknown and warrants future investigation.

MITOCHONDRIAL-TARGETED THERAPIES. Coenzyme Q10 is an electron transporter and a component of the mitochondrial ETC with antioxidant activities. Evidence from preclinical and clinical studies suggests that coenzyme Q10 has beneficial effects in HF.²⁶⁶ A meta-analysis reported that coenzyme Q10 supplementation in patients with HF increased exercise capacity and reduced mortality rate.²⁶⁷ However, poor bioavailability and the need for high doses to achieve mitochondrial accumulation limit its utility.

MitoQ is a mitochondrial-targeted antioxidant that readily accumulates in mitochondria, circumventing the limitation of coenzyme Q10 treatment. MitoQ treatment has been shown to enhance endothelial function in older patients,²⁶⁸ and improve cardiac function in preclinical models of HF.²⁶⁹ However, a recent pilot RCT reported MitoQ treatment did not affect myocardial or SkM energetics, LV function, or 6MWD in patients with dilated cardiomyopathy.²⁷⁰

HU6 is a metabolic accelerator that increases energy metabolism via its mitochondrial uncoupling activities. In HuMAIN-HFpEF (A novel controlled metabolic accelerator for the treatment of obesity-related heart failure with preserved ejection fraction: Rationale and design of the Phase 2a HuMAIN trial), HU6 treatment for 19 weeks resulted in modest weight loss among patients with obesity-HFpEF, with no significant change in peak VO_2 .²⁷¹

Elamipretide is a tetrapeptide with a high affinity for cardiolipin, a phospholipid primarily involved in the stabilization of ETC complexes.²⁷² In a canine model of HF, elamipretide reduced SkM ROS levels, prevented cardiolipin peroxidation, restored muscle fiber type composition, and improved mitochondrial oxidative capacity.²⁷³ However, in a phase 2 clinical trial involving patients with stable HFrEF, 4 weeks of subcutaneous elamipretide injections failed to improve LV ejection fraction or 6MWD.²⁷⁴ It was suggested that the short half-life of elamipretide may have led to the negative result.

Urolithin A, a microflora-derived metabolite and mitophagy activator, reduced inflammatory biomarkers, prevented accumulation of toxic lipid intermediates in mitochondria, and improved VO_2 -peak and 6MWD in middle-aged and older adults.^{275,276}

In summary, despite positive preclinical evidence, clinical translation of mitochondrial-targeted therapies to improve exercise capacity in HF has proven challenging.

ANTI-INFLAMMATORY THERAPIES. Anakinra, the IL-1 receptor antagonist, improved VO_2 -peak in patients with recently decompensated HFrEF in a pilot trial²⁷⁷; however, this positive result was not reproduced in the recently published phase 2 REDHART2 (REcently Decompensated Heart failure Anakinra Response Trial 2).²⁷⁸ Ziltivekimab, a monoclonal antibody targeting IL-6, is currently undergoing phase 3 clinical trials to assess its efficacy in improving symptoms, exercise capacity, and clinical outcomes in patients with HFpEF. Despite experimental evidence showing the beneficial effects of colchicine in HF, results from clinical trials have, thus far, been neutral.²⁷⁹ Ongoing trials are testing the efficacy of colchicine in patients with HFpEF (Table 2).

GDF-15 is a stress-induced cytokine that is elevated in response to inflammation, metabolic stress, and tissue injury. Anti-GDF-15 antibodies are investigated for their potential benefits in patients with cancer cachexia. Once such an antibody, ponsegromab improved appetite, increased body weight, and enhanced physical activity in patients with cancer

cachexia.²⁸⁰ Unfortunately, in the phase 3 GARDEN-TIMI 74 (Growth Differentiation Factor-15 Assessment of Rationale and Utility in Heart Failure evaluated by the Thrombolysis in Myocardial Infarction study group) trial, ponesegromab caused worsening of HF, leading to trial termination.

MODULATORS OF METABOLIC SIGNALING. Pharmacologic or genetic activation of AMP-activated protein kinase (AMPK), the master cellular energy sensor, is associated with enhanced SkM mitochondrial oxidative metabolism and biogenesis, improved insulin sensitivity, increased SkM regeneration, and improved exercise capacity in animal studies.²⁸¹⁻²⁸⁶ The direct AMPK activator, 5-aminoimidazole-4-carboxamide riboside, has poor bioavailability, limiting its utility. In patients with HFrEF, metformin, an indirect AMPK activator, did not improve exercise capacity despite increasing myocardial efficiency.²⁸⁷

Resveratrol is a natural polyphenol and an activator of AMPK. Resveratrol treatment improved exercise performance in animal models of HF.²⁸⁸ However, clinical evidence for the effectiveness of resveratrol in improving exercise capacity in HF is limited. One small study involving 30 patients with HFrEF reported that 3 months of daily resveratrol improved QoL and exercise capacity, as measured by the 6MWD.²⁸⁹ A pan-AMPK activator, ATX-304 (O304), and an isoform-selective AMPK activator, BLX-0871, are currently in early stages of development for their potential use in preserving lean mass during GLP-1 RA therapy following promising results in preclinical and pilot clinical studies.^{286,290}

SkM-generated BDNF is a cytokine highly expressed in oxidative myofibers, where it promotes SkM mitochondrial oxidative metabolism by activating AMPK and PGC-1 α .²⁹¹ Circulating BDNF levels are decreased in patients with HF and correlate with both exercise intolerance and poor prognosis. Subcutaneous injection of recombinant human BDNF improved exercise capacity in a mouse model of HF.²⁹²

Adiponectin, released from adipocytes, binds to AdipoR1 and AdipoR2 in SkM and other tissues, and initiates a signaling cascade leading to AMPK activation. The AdipoR1 and AdipoR2 agonist AdipoRon was shown to improve exercise capacity in aged and obese mice by activating AMPK, enhancing SkM oxidative metabolism, and reducing myosteatosis.²⁹³⁻²⁹⁵

Whether AMPK, BDNF, or adiponectin receptor-targeted therapies are beneficial in improving exercise capacity in HF remains to be elucidated.

Several other pharmacologic agents and nutritional interventions have been shown to improve

SkM dysfunction and exercise performance in experimental models but have limited clinical evidence supporting their use (Table 1).

CHALLENGES AND LIMITATIONS. It is important to note the limitations of translating results from preclinical models to patients in the context of HF. Most animal HF models do not accurately recapitulate clinical HF; this is particularly true for HFpEF. Interspecies differences in SkM and CV responses to stress also affect the translatability of the results from animal studies. These factors partly explain the discrepancy between preclinical and clinical study results regarding the efficacy of potential therapeutic interventions. Nevertheless, the insights gained from animal studies into the mechanisms and characteristics of HF-associated SkM dysfunction are invaluable for a deeper understanding of the disease and the discovery of novel therapies. In this regard, the slow translation of SkM-directed therapies with robust preclinical evidence remains a significant challenge.

A significant proportion of preclinical and clinical studies discussed in this review reported changes in SkM biology and structure, such as lean mass, but did not include measures of exercise capacity. Future studies should, when feasible, include measures of exercise capacity to evaluate the functional relevance of changes in SkM biology or lean mass. Recent advances in combined CPET and exercise-based noninvasive or invasive hemodynamic evaluations have enabled quantification of multisystem reserve capacities and identification of the predominant etiology of exercise intolerance in individual patients with HF. This is valuable in prioritizing specific therapeutic interventions targeted to the predominant deficit. However, such exercise-based tests are currently available only at a few expert centers. Finding ways to make such tests broadly available and readily accessible is a significant current challenge.

CONCLUSIONS AND FUTURE DIRECTIONS

Exercise intolerance is a cardinal symptom of HF, and SkM dysfunction is a key driver of exercise intolerance in HF. Both HFrEF and HFpEF are associated with significant SkM abnormalities, including reduced muscle mass, altered fiber-type composition, capillary rarefaction, and reduced mitochondrial content and oxidative capacity. These structural and metabolic SkM abnormalities have been shown to strongly correlate with exercise intolerance in HF. The growing recognition of SkM dysfunction as a major contributor to exercise intolerance should prompt further studies and a better understanding of

the phenomenon, thereby increasing the likelihood of developing novel therapies. According to current understanding, HF-associated oxidative stress, inflammation, and neurohormonal activation trigger an imbalance between protein degradation and synthesis, apoptosis, and mitochondrial dysfunction. Future studies that identify key molecular pathways linking HF to SkM dysfunction could pave the way for targeting these mechanisms via novel therapies.

Clinical studies have identified several molecules, including GDF-15, leptin, C-reactive protein, IL-6, and TNF- α , as potential biomarkers of SkM dysfunction in patients with HF.^{43,47,126,296,297} The application of these candidate molecules in clinical practice requires validation in large-scale clinical studies. Advances in omics technologies have enabled metabolic, transcriptomic, and proteomic profiling of plasma and SkM tissue to identify signatures of etiologic- or hemodynamic-based HF phenotypes and their correlations with exercise capacity. For instance, recent metabolomic profiling has revealed that the tryptophan metabolite kynurenine correlates with exercise capacity in HF.²⁹⁶ Future studies should harness the potential of integrating advanced omics techniques with gold-standard measures of exercise capacity to identify novel, clinically relevant biomarkers.

Exercise induces beneficial structural and functional changes in the SkM of patients with HF, which are associated with sustained improvements in exercise performance. A deeper understanding of the mechanisms underlying the salutary effects of exercise on SkM function would reveal promising targets for novel therapies to treat exercise intolerance in HF. In contrast to exercise, currently available HF

pharmacotherapies have not consistently improved exercise capacity, underscoring the urgent need for novel therapeutic modalities that yield robust, reliable gains in exercise and functional capacity. Emerging pharmacological and nutritional strategies targeting SkM oxidative stress, atrophy, and mitochondrial dysfunction have shown promise in improving SkM structure and function in preclinical HF models and in pilot clinical trials. Future research should focus on a deeper understanding of the mechanisms of action of these therapies, followed by testing their effectiveness in improving exercise tolerance in patients with HF in well-designed clinical trials.

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ADDRESS FOR CORRESPONDENCE: Dr Dinesh K. Kalra, Division of Cardiovascular Medicine, Department of Medicine, Lipid Clinic, Advanced Cardiac Imaging & Infiltrative Heart Disease Programs, University of Louisville School of Medicine, Rudd Heart & Lung Center, 201 Abraham Flexner Way, Suite 600, Louisville, Kentucky 40202, USA. E-mail: dinesh.kalra@louisville.edu. X handle: [@DineshKalra](https://twitter.com/DineshKalra).

REFERENCES

- Haykowsky MJ, Tomczak CR, Scott JM, Paterson DI, Kitzman DW. Determinants of exercise intolerance in patients with heart failure and reduced or preserved ejection fraction. *J Appl Physiol*. 2015;119:739–744.
- Pandey A, Shah SJ, Butler J, et al. Exercise intolerance in older adults with heart failure with preserved ejection fraction: JACC State-of-the-Art Review. *J Am Coll Cardiol*. 2021;78:1166–1187.
- Del Buono MG, Arena R, Borlaug BA, et al. Exercise intolerance in patients with heart failure: JACC state-of-the-art review. *J Am Coll Cardiol*. 2019;73:2209–2225.
- Piña IL, Apstein CS, Balady GJ, et al. Exercise and heart failure: a statement from the American Heart Association Committee on exercise, rehabilitation, and prevention. *Circulation*. 2003;107:1210–1225.
- Bozkurt B, Fonarow GC, Goldberg LR, et al. Cardiac rehabilitation for patients with heart failure: JACC expert panel. *J Am Coll Cardiol*. 2021;77:1454–1469.
- Kitzman DW, Whellan DJ, Duncan P, et al. Physical rehabilitation for older patients hospitalized for heart failure. *N Engl J Med*. 2021;385:203–216.
- von Haehling S, Arzt M, Doehner W, et al. Improving exercise capacity and quality of life using non-invasive heart failure treatments: evidence from clinical trials. *Eur J Heart Fail*. 2021;23:92–113.
- Esposito F, Mathieu-Costello O, Shabetai R, Wagner PD, Richardson RS. Limited maximal exercise capacity in patients with chronic heart failure: partitioning the contributors. *J Am Coll Cardiol*. 2010;55:1945–1954.
- Tucker WJ, Haykowsky MJ, Seo Y, Stehling E, Forman DE. Impaired exercise tolerance in heart failure: role of skeletal muscle morphology and function. *Curr Heart Fail Rep*. 2018;15:323–331.
- Houstis NE, Eisman AS, Pappagianopoulos PP, et al. Exercise intolerance in heart failure with preserved ejection fraction: diagnosing and ranking its causes using personalized O. *Circulation*. 2018;137:148–161.
- Kitzman DW, Nicklas B, Kraus WE, et al. Skeletal muscle abnormalities and exercise intolerance in older patients with heart failure and preserved ejection fraction. *Am J Physiol Heart Circ Physiol*. 2014;306:H1364–H1370.
- Molina AJ, Bharadwaj MS, Van Horn C, et al. Skeletal muscle mitochondrial content, oxidative capacity, and Mfn2 expression are reduced in older patients with heart failure and preserved ejection fraction and are related to exercise intolerance. *JACC Heart Fail*. 2016;4:636–645.
- Scandalis L, Kitzman DW, Nicklas BJ, et al. Skeletal muscle mitochondrial respiration and exercise intolerance in patients with heart failure with preserved ejection fraction. *JAMA Cardiol*. 2023;8:575–584.

14. Wilson JR, Mancini DM, Dunkman WB. Exertional fatigue due to skeletal muscle dysfunction in patients with heart failure. *Circulation*. 1993;87:470-475.
15. Minotti JR, Christoph I, Oka R, Weiner MW, Wells L, Massie BM. Impaired skeletal muscle function in patients with congestive heart failure. Relationship to systemic exercise performance. *J Clin Invest*. 1991;88:2077-2082.
16. Mancini DM, Walter G, Reichel N, et al. Contribution of skeletal muscle atrophy to exercise intolerance and altered muscle metabolism in heart failure. *Circulation*. 1992;85:1364-1373.
17. Borlaug BA, Melenovsky V, Russell SD, et al. Impaired chronotropic and vasodilator reserves limit exercise capacity in patients with heart failure and a preserved ejection fraction. *Circulation*. 2006;114:2138-2147.
18. Mancini DM, Eisen H, Kussmaul W, Mull R, Edmunds LH, Wilson JR. Value of peak exercise oxygen consumption for optimal timing of cardiac transplantation in ambulatory patients with heart failure. *Circulation*. 1991;83:778-786.
19. Lewis GD, Tada A, Landsteiner I, Borlaug BA. Physiologic phenotyping of responses to exercise and activity in heart failure. *Circ Res*. 2025;137:290-315.
20. Gitt AK, Wasserman K, Kilkowski C, et al. Exercise anaerobic threshold and ventilatory efficiency identify heart failure patients for high risk of early death. *Circulation*. 2002;106:3079-3084.
21. Arena R, Myers J, Abella J, et al. Development of a ventilatory classification system in patients with heart failure. *Circulation*. 2007;115:2410-2417.
22. Koo BK. Assessment of muscle quantity, quality and function. *J Obes Metab Syndr*. 2022;31:9-16.
23. Casey P, Alasmir M, McLaughlin J, et al. The current use of ultrasound to measure skeletal muscle and its ability to predict clinical outcomes: a systematic review. *J Cachexia Sarcopenia Muscle*. 2022;13:2298-2309.
24. Tang T, Xie L, Hu S, et al. Serum creatinine and cystatin C-based diagnostic indices for sarcopenia in advanced non-small cell lung cancer. *J Cachexia Sarcopenia Muscle*. 2022;13:1800-1810.
25. Haykowsky MJ, Brubaker PH, John JM, Stewart KP, Morgan TM, Kitzman DW. Determinants of exercise intolerance in elderly heart failure patients with preserved ejection fraction. *J Am Coll Cardiol*. 2011;58:265-274.
26. Dhakal BP, Malhotra R, Murphy RM, et al. Mechanisms of exercise intolerance in heart failure with preserved ejection fraction: the role of abnormal peripheral oxygen extraction. *Circ Heart Fail*. 2015;8:286-294.
27. Borlaug BA, Reddy YNV, Braun A, et al. Cardiac and metabolic effects of dapagliflozin in heart failure with preserved ejection fraction: the CAMEO-DAPA trial. *Circulation*. 2023;148:834-844.
28. Sarma S, MacNamara JP, Balmain BN, et al. Challenging the hemodynamic hypothesis in heart failure with preserved ejection fraction: is exercise capacity limited by elevated pulmonary capillary wedge pressure? *Circulation*. 2023;147:378-387.
29. Borlaug BA, Anstrom KJ, Lewis GD, et al. Effect of inorganic nitrite vs placebo on exercise capacity among patients with heart failure with preserved ejection fraction: the INDIE-HFpEF randomized clinical trial. *JAMA*. 2018;320:1764-1773.
30. Borlaug BA, Koepp KE, Reddy YNV, et al. Inorganic nitrite to amplify the benefits and tolerability of exercise training in heart failure with preserved ejection fraction: the INABLE-training trial. *Mayo Clin Proc*. 2024;99:206-217.
31. Shah SJ, Borlaug BA, Chung ES, et al. Atrial shunt device for heart failure with preserved and mildly reduced ejection fraction (REDUCE LAP-HF II): a randomised, multicentre, blinded, sham-controlled trial. *Lancet*. 2022;399:1130-1140.
32. Franciosa JA, Cohn JN. Effect of isosorbide dinitrate on response to submaximal and maximal exercise in patients with congestive heart failure. *Am J Cardiol*. 1979;43:1009-1014.
33. Sachdev V, Sharma K, Keteyian SJ, et al. Supervised exercise training for chronic heart failure with preserved ejection fraction: a scientific statement from the American Heart Association and American College of Cardiology. *J Am Coll Cardiol*. 2023;81:1524-1542.
34. Pandey A, Parashar A, Kumbhani D, et al. Exercise training in patients with heart failure and preserved ejection fraction: meta-analysis of randomized control trials. *Circ Heart Fail*. 2015;8:33-40.
35. Haykowsky MJ, Brubaker PH, Stewart KP, Morgan TM, Eggebeen J, Kitzman DW. Effect of endurance training on the determinants of peak exercise oxygen consumption in elderly patients with stable compensated heart failure and preserved ejection fraction. *J Am Coll Cardiol*. 2012;60:120-128.
36. Fu TC, Yang NI, Wang CH, et al. Aerobic interval training elicits different hemodynamic adaptations between heart failure patients with preserved and reduced ejection fraction. *Am J Phys Med Rehabil*. 2016;95:15-27.
37. Rustad LA, Nytrøen K, Amundsen BH, Gullestad L, Aakhus S. One year of high-intensity interval training improves exercise capacity, but not left ventricular function in stable heart transplant recipients: a randomised controlled trial. *Eur J Prev Cardiol*. 2014;21:181-191.
38. Witte KK, Nikitin NP, De Silva R, Cleland JG, Clark AL. Exercise capacity and cardiac function assessed by tissue Doppler imaging in chronic heart failure. *Heart*. 2004;90:1144-1150.
39. Wilson JR. Exercise intolerance in heart failure. Importance of skeletal muscle. *Circulation*. 1995;91:559-561.
40. Weiss K, Schär M, Panjath GS, et al. Fatigability, exercise intolerance, and abnormal skeletal muscle energetics in heart failure. *Circ Heart Fail*. 2017;10:e004129.
41. Bowen TS, Rolim NP, Fischer T, et al. Heart failure with preserved ejection fraction induces molecular, mitochondrial, histological, and functional alterations in rat respiratory and limb skeletal muscle. *Eur J Heart Fail*. 2015;17:263-272.
42. Drexler H, Riede U, Münzel T, König H, Funke E, Just H. Alterations of skeletal muscle in chronic heart failure. *Circulation*. 1992;85:1751-1759.
43. Bekfani T, Bekhite Elsaied M, Derlien S, et al. Skeletal muscle function, structure, and metabolism in patients with heart failure with reduced ejection fraction and heart failure with preserved ejection fraction. *Circ Heart Fail*. 2020;13:e007198.
44. Mancini DM, Coyle E, Coggan A, et al. Contribution of intrinsic skeletal muscle changes to 31P NMR skeletal muscle metabolic abnormalities in patients with chronic heart failure. *Circulation*. 1989;80:1338-1346.
45. Duscha BD, Kraus WE, Keteyian SJ, et al. Capillary density of skeletal muscle: a contributing mechanism for exercise intolerance in class II-III chronic heart failure independent of other peripheral alterations. *J Am Coll Cardiol*. 1999;33:1956-1963.
46. Kitzman DW, Brubaker PH, Morgan TM, Stewart KP, Little WC. Exercise training in older patients with heart failure and preserved ejection fraction: a randomized, controlled, single-blind trial. *Circ Heart Fail*. 2010;3:659-667.
47. Fülster S, Tacke M, Sandek A, et al. Muscle wasting in patients with chronic heart failure: results from the studies investigating comorbidities aggravating heart failure (SICA-HF). *Eur Heart J*. 2013;34:512-519.
48. Anker SD, Ponikowski P, Varney S, et al. Wasting as independent risk factor for mortality in chronic heart failure. *Lancet*. 1997;349:1050-1053.
49. Miller MS, Vanburen P, Lewinter MM, et al. Mechanisms underlying skeletal muscle weakness in human heart failure: alterations in single fiber myosin protein content and function. *Circ Heart Fail*. 2009;2:700-706.
50. Miller MS, VanBuren P, LeWinter MM, et al. Chronic heart failure decreases cross-bridge kinetics in single skeletal muscle fibres from humans. *J Physiol*. 2010;588:4039-4053.
51. von Haehling S, Ebner N, Dos Santos MR, Springer J, Anker SD. Muscle wasting and cachexia in heart failure: mechanisms and therapies. *Nat Rev Cardiol*. 2017;14:323-341.
52. Hambrecht R, Schulze PC, Gielen S, et al. Reduction of insulin-like growth factor-I expression in the skeletal muscle of noncachectic patients with chronic heart failure. *J Am Coll Cardiol*. 2002;39:1175-1181.
53. Toth MJ, Ward K, van der Velden J, et al. Chronic heart failure reduces Akt phosphorylation in human skeletal muscle: relationship to muscle size and function. *J Appl Physiol*. 2011;110:892-900.
54. Schulze PC, Fang J, Kassik KA, et al. Transgenic overexpression of locally acting insulin-like growth factor-1 inhibits ubiquitin-mediated muscle atrophy in chronic left-ventricular dysfunction. *Circ Res*. 2005;97:418-426.
55. Adams V, Wunderlich S, Mangner N, et al. Ubiquitin-proteasome-system and enzymes of energy metabolism in skeletal muscle of patients with HFpEF and HFrEF. *ESC Heart Fail*. 2021;8:2556-2568.
56. Gielen S, Sandri M, Kozarek I, et al. Exercise training attenuates MuRF-1 expression in the

skeletal muscle of patients with chronic heart failure independent of age: the randomized Leipzig exercise intervention in chronic heart failure and aging catabolism study. *Circulation*. 2012;125:2716-2727.

57. Garnham JO, Roberts LD, Caspi T, et al. Divergent skeletal muscle mitochondrial phenotype between male and female patients with chronic heart failure. *J Cachexia Sarcopenia Muscle*. 2020;11:79-88.

58. Dalla LL, Ravara B, Angelini A, et al. Beneficial effects on skeletal muscle of the angiotensin II type 1 receptor blocker irbesartan in experimental heart failure. *Circulation*. 2001;103:2195-2200.

59. Adams V, Jiang H, Yu J, et al. Apoptosis in skeletal myocytes of patients with chronic heart failure is associated with exercise intolerance. *J Am Coll Cardiol*. 1999;33:959-965.

60. Nambu H, Takada S, Maekawa S, et al. Inhibition of xanthine oxidase in the acute phase of myocardial infarction prevents skeletal muscle abnormalities and exercise intolerance. *Cardiovasc Res*. 2021;117:805-819.

61. Espino-Gonzalez E, Tickle PG, Altara R, et al. Caloric restriction rejuvenates skeletal muscle growth in heart failure with preserved ejection fraction. *JACC Basic Transl Sci*. 2024;9:223-240.

62. Sente T, Van Berendonck AM, Jonckheere AI, et al. Primary skeletal muscle myoblasts from chronic heart failure patients exhibit loss of anti-inflammatory and proliferative activity. *BMC Cardiovasc Disord*. 2016;16:107.

63. Yoshida T, Galvez S, Tiwari S, et al. Angiotensin II inhibits satellite cell proliferation and prevents skeletal muscle regeneration. *J Biol Chem*. 2013;288:23823-23832.

64. Sullivan MJ, Green HJ, Cobb FR. Skeletal muscle biochemistry and histology in ambulatory patients with long-term heart failure. *Circulation*. 1990;81:518-527.

65. Bowen TS, Herz C, Rolim NPL, et al. Effects of endurance training on detrimental structural, cellular, and functional alterations in skeletal muscles of heart failure with preserved ejection fraction. *J Card Fail*. 2018;24:603-613.

66. Gortan Cappellari G, Aleksova A, Dal Ferro M, et al. 3 PUFA-enriched diet preserves skeletal muscle mitochondrial function and redox state and prevents muscle mass loss in mice with chronic heart failure. *Nutrients*. 2023;15:310.

67. Haykowsky MJ, Nicklas BJ, Brubaker PH, et al. Regional adipose distribution and its relationship to exercise intolerance in older obese patients who have heart failure with preserved ejection fraction. *JACC Heart Fail*. 2018;6:640-649.

68. Haykowsky MJ, Kouba EJ, Brubaker PH, Nicklas BJ, Eggebeen J, Kitzman DW. Skeletal muscle composition and its relation to exercise intolerance in older patients with heart failure and preserved ejection fraction. *Am J Cardiol*. 2014;113:1211-1216.

69. Larsen AI, Lindal S, Aukrust P, Toft I, Aarsland T, Dickstein K. Effect of exercise training on skeletal muscle fibre characteristics in men with chronic heart failure. Correlation between

skeletal muscle alterations, cytokines and exercise capacity. *Int J Cardiol*. 2002;83:25-32.

70. Borlaug BA, Olson TP, Lam CS, et al. Global cardiovascular reserve dysfunction in heart failure with preserved ejection fraction. *J Am Coll Cardiol*. 2010;56:845-854.

71. Manetos C, Dimopoulos S, Tzanis G, et al. Skeletal muscle microcirculatory abnormalities are associated with exercise intolerance, ventilatory inefficiency, and impaired autonomic control in heart failure. *J Heart Lung Transplant*. 2011;30:1403-1408.

72. Dos Santos MR, Saitoh M, Ebner N, et al. Sarcopenia and endothelial function in patients with chronic heart failure: results from the studies investigating comorbidities aggravating heart failure (SICA-HF). *J Am Med Dir Assoc*. 2017;18:240-245.

73. Bauer TA, Reusch JE, Levi M, Regensteiner JG. Skeletal muscle deoxygenation after the onset of moderate exercise suggests slowed microvascular blood flow kinetics in type 2 diabetes. *Diabetes Care*. 2007;30:2880-2885.

74. Erbs S, Höllriegel R, Linke A, et al. Exercise training in patients with advanced chronic heart failure (NYHA IIIb) promotes restoration of peripheral vasomotor function, induction of endogenous regeneration, and improvement of left ventricular function. *Circ Heart Fail*. 2010;3:486-494.

75. Hambrecht R, Fiehn E, Yu J, et al. Effects of endurance training on mitochondrial ultrastructure and fiber type distribution in skeletal muscle of patients with stable chronic heart failure. *J Am Coll Cardiol*. 1997;29:1067-1073.

76. Spangenburg EE, Talmadge RJ, Musch TI, Pfeifer PC, McAllister RM, Williams JH. Changes in skeletal muscle myosin heavy chain isoform content during congestive heart failure. *Eur J Appl Physiol*. 2002;87:182-186.

77. Simonini A, Long CS, Dudley GA, Yue P, McElhinny J, Massie BM. Heart failure in rats causes changes in skeletal muscle morphology and gene expression that are not explained by reduced activity. *Circ Res*. 1996;79:128-136.

78. Massie BM, Conway M, Yonge R, et al. 31P nuclear magnetic resonance evidence of abnormal skeletal muscle metabolism in patients with congestive heart failure. *Am J Cardiol*. 1987;60:309-315.

79. Schaufelberger M, Eriksson BO, Grimby G, Held P, Swedberg K. Skeletal muscle alterations in patients with chronic heart failure. *Eur Heart J*. 1997;18:971-980.

80. Nijholt KT, Sánchez-Aguilera PI, Voorrips SN, de Boer RA, Westenbrink BD. Exercise: a molecular tool to boost muscle growth and mitochondrial performance in heart failure? *Eur J Heart Fail*. 2022;24:287-298.

81. Vescovo G, Serafini F, Dalla Libera L, et al. Skeletal muscle myosin heavy chains in heart failure: correlation between magnitude of the isozyme shift, exercise capacity, and gas exchange measurements. *Am Heart J*. 1998;135:130-137.

82. Murphy RM, Watt MJ, Febbraio MA. Metabolic communication during exercise. *Nat Metab*. 2020;2:805-816.

83. Coen PM, Jubrias SA, Distefano G, et al. Skeletal muscle mitochondrial energetics are associated with maximal aerobic capacity and walking speed in older adults. *J Gerontol A Biol Sci Med Sci*. 2013;68:447-455.

84. Leermakers PA, Gosker HR. Skeletal muscle mitophagy in chronic disease: implications for muscle oxidative capacity? *Curr Opin Clin Nutr Metab Care*. 2016;19:427-433.

85. Ito A, Hashimoto M, Tanihata J, et al. Involvement of Parkin-mediated mitophagy in the pathogenesis of chronic obstructive pulmonary disease-related sarcopenia. *J Cachexia Sarcopenia Muscle*. 2022;13:1864-1882.

86. Magnusson G, Kaijser L, Rong H, Isberg B, Sylvén C, Saltin B. Exercise capacity in heart failure patients: relative importance of heart and skeletal muscle. *Clin Physiol*. 1996;16:183-195.

87. Schrepper A, Schwarzer M, Schöpe M, Amorim PA, Doenst T. Biphasic response of skeletal muscle mitochondria to chronic cardiac pressure overload - role of respiratory chain complex activity. *J Mol Cell Cardiol*. 2012;52:125-135.

88. Garnier A, Fortin D, Deloménie C, Momken I, Veksler V, Ventura-Clapier R. Depressed mitochondrial transcription factors and oxidative capacity in rat failing cardiac and skeletal muscles. *J Physiol*. 2003;551:491-501.

89. Rosca MG, Okere IA, Sharma N, Stanley WC, Recchia FA, Hoppel CL. Altered expression of the adenine nucleotide translocase isoforms and decreased ATP synthase activity in skeletal muscle mitochondria in heart failure. *J Mol Cell Cardiol*. 2009;46:927-935.

90. Zamani P, Proto EA, Wilson N, et al. Multi-modality assessment of heart failure with preserved ejection fraction skeletal muscle reveals differences in the machinery of energy fuel metabolism. *ESC Heart Fail*. 2021;8:2698-2712.

91. Knuiman P, Straw S, Gierula J, et al. Quantifying the relationship and contribution of mitochondrial respiration to systemic exercise limitation in heart failure. *ESC Heart Fail*. 2021;8:898-907.

92. Caspi T, Straw S, Cheng C, et al. Unique transcriptome signature distinguishes patients with heart failure with myopathy. *J Am Heart Assoc*. 2020;9:e017091.

93. Sullivan MJ, Green HJ, Cobb FR. Altered skeletal muscle metabolic response to exercise in chronic heart failure. Relation to skeletal muscle aerobic enzyme activity. *Circulation*. 1991;84:1597-1607.

94. van der Ent M, Jeneson JA, Remme WJ, Berger R, Ciampicotti R, Visser F. A non-invasive selective assessment of type I fibre mitochondrial function using 31P NMR spectroscopy. Evidence for impaired oxidative phosphorylation rate in skeletal muscle in patients with chronic heart failure. *Eur Heart J*. 1998;19:124-131.

95. Doehner W, Turhan G, Leyva F, et al. Skeletal muscle weakness is related to insulin resistance in patients with chronic heart failure. *ESC Heart Fail*. 2015;2:85-89.

96. Swan JW, Anker SD, Walton C, et al. Insulin resistance in chronic heart failure: relation to

severity and etiology of heart failure. *J Am Coll Cardiol*. 1997;30:527-532.

97. Ohta Y, Kinugawa S, Matsushima S, et al. Oxidative stress impairs insulin signal in skeletal muscle and causes insulin resistance in postinfarct heart failure. *Am J Physiol Heart Circ Physiol*. 2011;300:H1637-H1644.

98. Doehner W, Gathercole D, Ciccoira M, et al. Reduced glucose transporter GLUT4 in skeletal muscle predicts insulin resistance in non-diabetic chronic heart failure patients independently of body composition. *Int J Cardiol*. 2010;138:19-24.

99. Kemppainen J, Tsuchida H, Stolen K, et al. Insulin signalling and resistance in patients with chronic heart failure. *J Physiol*. 2003;550:305-315.

100. Velasquez FC, Roman B, Hernández-Ochoa EO, et al. Contribution of skeletal muscle-specific microRNA-133b to insulin resistance in heart failure. *Am J Physiol Heart Circ Physiol*. 2023;324:H598-H609.

101. Mettauer B, Zoll J, Sanchez H, et al. Oxidative capacity of skeletal muscle in heart failure patients versus sedentary or active control subjects. *J Am Coll Cardiol*. 2001;38:947-954.

102. De Sousa E, Veksler V, Bigard X, Mateo P, Ventura-Clapier R. Heart failure affects mitochondrial but not myofibrillar intrinsic properties of skeletal muscle. *Circulation*. 2000;102:1847-1853.

103. Lewsey SC, Samuel TJ, Schär M, et al. Skeletal muscle quantity versus quality in heart failure: exercise intolerance and outcomes in older patients with HFpEF are related to abnormal skeletal muscle metabolism rather than age-related skeletal muscle loss. *Circ Heart Fail*. 2025;18:e012512.

104. Powers SK, Morton AB, Hyatt H, Hinkley MJ. The renin-angiotensin system and skeletal muscle. *Exerc Sport Sci Rev*. 2018;46:205-214.

105. Inoue N, Kinugawa S, Suga T, et al. Angiotensin II-induced reduction in exercise capacity is associated with increased oxidative stress in skeletal muscle. *Am J Physiol Heart Circ Physiol*. 2012;302:H1202-H1210.

106. Liu Y, Bi X, Zhang Y, Wang Y, Ding W. Mitochondrial dysfunction/NLRP3 inflammasome axis contributes to angiotensin II-induced skeletal muscle wasting via PPAR- γ . *Lab Invest*. 2020;100:712-726.

107. Du Bois P, Pablo Tortola C, Lodka D, et al. Angiotensin II induces skeletal muscle atrophy by activating TFEB-mediated MuRF1 expression. *Circ Res*. 2015;117:424-436.

108. Cozzoli A, Liantonio A, Conte E, et al. Angiotensin II modulates mouse skeletal muscle resting conductance to chloride and potassium ions and calcium homeostasis via the AT1 receptor and NADPH oxidase. *Am J Physiol Cell Physiol*. 2014;307:C634-C647.

109. Guzmán MG, Báez AL, Lo Presti MS, et al. Functional and structural alterations of cardiac and skeletal muscle mitochondria in heart failure patients. *Arch Med Res*. 2014;45:237-246.

110. Lv J, Li Y, Shi S, et al. Skeletal muscle mitochondrial remodeling in heart failure: an update on mechanisms and therapeutic opportunities. *Biomed Pharmacother*. 2022;155:113833.

111. Masunaga T, Suenaga T, Matsushima S, et al. Reduction in acetylation of superoxide dismutase 2 in skeletal muscle improves exercise capacity in mice with heart failure. *J Cachexia Sarcopenia Muscle*. 2025;16:e13850.

112. Tsuda M, Fukushima A, Matsumoto J, et al. Protein acetylation in skeletal muscle mitochondria is involved in impaired fatty acid oxidation and exercise intolerance in heart failure. *J Cachexia Sarcopenia Muscle*. 2018;9:844-859.

113. Jing E, Emanuelli B, Hirshey MD, et al. Sirtuin-3 (Sirt3) regulates skeletal muscle metabolism and insulin signaling via altered mitochondrial oxidation and reactive oxygen species production. *Proc Natl Acad Sci U S A*. 2011;108:14608-14613.

114. Macconi D, Perico L, Longaretti L, et al. Sirtuin3 dysfunction is the key determinant of skeletal muscle insulin resistance by angiotensin II. *PLoS One*. 2015;10:e0127172.

115. Baba SP, Zhang D, Singh M, et al. Deficiency of aldose reductase exacerbates early pressure overload-induced cardiac dysfunction and autophagy in mice. *J Mol Cell Cardiol*. 2018;118:183-192.

116. Barrera G, Pizzimenti S, Daga M, et al. Lipid peroxidation-derived aldehydes, 4-hydroxynonenal and malondialdehyde in aging-related disorders. *Antioxidants (Basel)*. 2018;7:102.

117. Srivastava S, Watowich SJ, Petrash JM, Srivastava SK, Bhatnagar A. Structural and kinetic determinants of aldehyde reduction by aldose reductase. *Biochemistry*. 1999;38:42-54.

118. Blacquart L, Baba SP, Kwiatkowski S, et al. Carnosine and anserine homeostasis in skeletal muscle and heart is controlled by β -alanine transamination. *J Physiol*. 2016;594:4849-4863.

119. Zhao J, Conklin DJ, Guo Y, et al. Cardiospecific overexpression of ATPGD1 (carnosine synthase) increases histidine dipeptide levels and prevents myocardial ischemia reperfusion injury. *J Am Heart Assoc*. 2020;9:e015222.

120. Chaudhari M, Zelko I, Lorkiewicz P, et al. Metabolic pathways for removing reactive aldehydes are diminished in the skeletal muscle during heart failure. *Skelet Muscle*. 2024;14:24.

121. Lavine KJ, Sierra OL. Skeletal muscle inflammation and atrophy in heart failure. *Heart Fail Rev*. 2017;22:179-189.

122. Gielen S, Adams V, Möbius-Winkler S, et al. Anti-inflammatory effects of exercise training in the skeletal muscle of patients with chronic heart failure. *J Am Coll Cardiol*. 2003;42:861-868.

123. Deswal A, Petersen NJ, Feldman AM, Young JB, White BG, Mann DL. Cytokines and cytokine receptors in advanced heart failure: an analysis of the cytokine database from the Vesnarinone trial (VEST). *Circulation*. 2001;103:2055-2059.

124. Murphy SP, Kakkar R, McCarthy CP, Januzzi JL. Inflammation in heart failure: JACC state-of-the-art review. *J Am Coll Cardiol*. 2020;75:1324-1340.

125. Zhou J, Liu B, Liang C, Li Y, Song YH. Cytokine signaling in skeletal muscle wasting. *Trends Endocrinol Metab*. 2016;27:335-347.

126. Yuzefpolskaya M, Bohn B, Ladanyi A, et al. Alterations in the sarcopenia index are associated

with inflammation, gut, and oral microbiota among heart failure, left ventricular assist device, and heart transplant patients. *J Heart Lung Transplant*. 2024;43:1395-1408.

127. Hambrecht R, Adams V, Gielen S, et al. Exercise intolerance in patients with chronic heart failure and increased expression of inducible nitric oxide synthase in the skeletal muscle. *J Am Coll Cardiol*. 1999;33:174-179.

128. Torres SH, De Sanctis JB, de L Briceño M, Hernández N, Finol HJ. Inflammation and nitric oxide production in skeletal muscle of type 2 diabetic patients. *J Endocrinol*. 2004;181:419-427.

129. Sadek J, Hall DT, Colalillo B, et al. Pharmacological or genetic inhibition of iNOS prevents cachexia-mediated muscle wasting and its associated metabolism defects. *EMBO Mol Med*. 2021;13:e13591.

130. Schmid V, Foulkes SJ, Walesiak D, et al. Impact of whole-body and skeletal muscle composition on peak oxygen uptake in heart failure: a systematic review and meta-analysis. *Eur Heart J Open*. 2024;4:oeae082.

131. Foulkes SJ, Moura-Ferreira S, Milani M, et al. Mechanisms of exercise limitation in heart failure with preserved ejection fraction and obesity: a case of engine-chassis mismatch. *Am J Physiol Heart Circ Physiol*. 2025;329:H1126-H1136.

132. Pandit A, Gupta M, Arabie DA, et al. Obesity and heart failure with preserved ejection fraction: a clinical nexus for exercise intolerance. *Obesity (Silver Spring)*. 2026;34:984-996.

133. Packer M. The adipokine hypothesis of heart failure with a preserved ejection fraction: a novel framework to explain pathogenesis and guide treatment. *J Am Coll Cardiol*. 2025;86:1269-1373.

134. Matsue Y, Kamiya K, Saito H, et al. Prevalence and prognostic impact of the coexistence of multiple frailty domains in elderly patients with heart failure: the FRAGILE-HF cohort study. *Eur J Heart Fail*. 2020;22:2112-2119.

135. von Haehling S, Garfias Macedo T, Valentova M, et al. Muscle wasting as an independent predictor of survival in patients with chronic heart failure. *J Cachexia Sarcopenia Muscle*. 2020;11:1242-1249.

136. Konishi M, Kagiya N, Kamiya K, et al. Impact of sarcopenia on prognosis in patients with heart failure with reduced and preserved ejection fraction. *Eur J Prev Cardiol*. 2021;28:1022-1029.

137. Shrauner W, Lord EM, Nguyen XT, et al. Frailty and cardiovascular mortality in more than 3 million US veterans. *Eur Heart J*. 2022;43:818-826.

138. Sato R, Vatic M, Peixoto da Fonseca GW, Anker SD, von Haehling S. Biological basis and treatment of frailty and sarcopenia. *Cardiovasc Res*. 2024;120:982-998.

139. Tezze C, Romanello V, Desbats MA, et al. Age-associated loss of OPA1 in muscle impacts muscle mass, metabolic homeostasis, systemic inflammation, and epithelial senescence. *Cell Metab*. 2017;25:1374-1389.e6.

140. Lau ES, Cunningham T, Hardin KM, et al. Sex differences in cardiometabolic traits and determinants of exercise capacity in heart failure

with preserved ejection fraction. *JAMA Cardiol.* 2020;5:30–37.

141. Hearon CM, Skow RJ, Wakeham DJ, et al. Sex differences in heart failure with preserved ejection fraction: pathophysiology: a detailed invasive hemodynamic and echocardiographic analysis. *Am J Physiol Heart Circ Physiol.* 2026;331:H11–H22.

142. Beale AL, Nanayakkara S, Segal L, et al. Sex differences in heart failure with preserved ejection fraction pathophysiology: a detailed invasive hemodynamic and echocardiographic analysis. *JACC Heart Fail.* 2019;7:239–249.

143. Skow RJ, Foulkes SJ, Wang J, et al. Sex-based differences in peak oxygen uptake among individuals with heart failure: systematic review and meta-analysis. *J Appl Physiol.* 2025;139:45–57.

144. Chyou JY, Qin H, Butler J, Voors AA, Lam CSP. Sex-related similarities and differences in responses to heart failure therapies. *Nat Rev Cardiol.* 2024;21:498–516.

145. Sorimachi H, Obokata M, Takahashi N, et al. Pathophysiologic importance of visceral adipose tissue in women with heart failure and preserved ejection fraction. *Eur Heart J.* 2021;42:1595–1605.

146. Sorimachi H, Omote K, Omar M, et al. Sex and central obesity in heart failure with preserved ejection fraction. *Eur J Heart Fail.* 2022;24:1359–1370.

147. Torres MJ, Kew KA, Ryan TE, et al. 17 β -Estradiol directly lowers mitochondrial membrane microviscosity and improves bioenergetic function in skeletal muscle. *Cell Metab.* 2018;27:167–179.e7.

148. Knowlton AA, Korzick DH. Estrogen and the female heart. *Mol Cell Endocrinol.* 2014;389:31–39.

149. Greendale GA, Sternfeld B, Huang M, et al. Changes in body composition and weight during the menopause transition. *JCI Insight.* 2019;4:e124865.

150. Wood N, Critchlow A, Cheng CW, et al. Sex differences in skeletal muscle pathology in patients with heart failure and reduced ejection fraction. *Circ Heart Fail.* 2024;17:e011471.

151. Landsteiner I, Verwerf J, Belov D, Verbrugge FH, Lewis GD. Heart failure and preserved ejection fraction: pathophysiology, clinical assessment, and management of exercise intolerance. *Eur Heart J.* 2026;47:3065–3101.

152. Gao M, Bhatia K, Kapoor A, et al. SGLT2 inhibitors, functional capacity, and quality of life in patients with heart failure: a systematic review and meta-analysis. *JAMA Netw Open.* 2024;7:e245135.

153. Tanashat M, Manasrah A, Abouzid M. Effects of dapagliflozin and empagliflozin on 6-min walk distance in heart failure with preserved and reduced ejection fraction: a systematic review and meta-analysis of randomized controlled trials involving 2624 patients. *Eur J Clin Pharmacol.* 2024;80:951–963.

154. Ziesche S, Cobb FR, Cohn JN, Johnson G, Tristani F. Hydralazine and isosorbide dinitrate combination improves exercise tolerance in heart failure. Results from V-HeFT I and V-HeFT II. The

V-HeFT VA Cooperative Studies Group. *Circulation.* 1993;87:VI56–VI64.

155. Lorenzo M, Jacobs-Cachá C, Palau P, et al. Short-term changes in peak VO. *JACC Heart Fail.* 2023;11:1611–1622.

156. Mueller S, Winzer EB, Duvinage A, et al. Effect of high-intensity interval training, moderate continuous training, or guideline-based physical activity advice on peak oxygen consumption in patients with heart failure with preserved ejection fraction: a randomized clinical trial. *JAMA.* 2021;325:542–551.

157. Kitzman DW, Brubaker P, Morgan T, et al. Effect of caloric restriction or aerobic exercise training on peak oxygen consumption and quality of life in obese older patients with heart failure with preserved ejection fraction: a randomized clinical trial. *JAMA.* 2016;315:36–46.

158. O'Connor CM, Whellan DJ, Lee KL, et al. Efficacy and safety of exercise training in patients with chronic heart failure: HF-ACTION randomized controlled trial. *JAMA.* 2009;301:1439–1450.

159. Fleg JL, Cooper LS, Borlaug BA, et al. Exercise training as therapy for heart failure: current status and future directions. *Circ Heart Fail.* 2015;8:209–220.

160. Flynn KE, Piña IL, Whellan DJ, et al. Effects of exercise training on health status in patients with chronic heart failure: HF-ACTION randomized controlled trial. *JAMA.* 2009;301:1451–1459.

161. Hambrecht R, Niebauer J, Fiehn E, et al. Physical training in patients with stable chronic heart failure: effects on cardiorespiratory fitness and ultrastructural abnormalities of leg muscles. *J Am Coll Cardiol.* 1995;25:1239–1249.

162. Hambrecht R, Fiehn E, Weigl C, et al. Regular physical exercise corrects endothelial dysfunction and improves exercise capacity in patients with chronic heart failure. *Circulation.* 1998;98:2709–2715.

163. Tryfonos A, Tzanis G, Pitsolis T, et al. Exercise training enhances angiogenesis-related gene responses in skeletal muscle of patients with chronic heart failure. *Cells.* 2021;10:1915.

164. Linke A, Adams V, Schulze PC, et al. Antioxidative effects of exercise training in patients with chronic heart failure: increase in radical scavenger enzyme activity in skeletal muscle. *Circulation.* 2005;111:1763–1770.

165. Cai M, Wang Q, Liu Z, Jia D, Feng R, Tian Z. Effects of different types of exercise on skeletal muscle atrophy, antioxidant capacity and growth factors expression following myocardial infarction. *Life Sci.* 2018;213:40–49.

166. Bacurau AV, Jannig PR, de Moraes WM, et al. Akt/mTOR pathway contributes to skeletal muscle anti-atrophic effect of aerobic exercise training in heart failure mice. *Int J Cardiol.* 2016;214:137–147.

167. Bardi E, Majerczak J, Zoladz JA, et al. Voluntary physical activity counteracts chronic heart failure progression affecting both cardiac function and skeletal muscle in the transgenic Tg α q*44 mouse model. *Physiol Rep.* 2019;7:e14161.

168. Moreira JB, Bechara LR, Bozi LH, et al. High-versus moderate-intensity aerobic exercise training effects on skeletal muscle of infarcted rats. *J Appl Physiol.* 2013;114:1029–1041.

169. Quiariarte H, Noland RC, Stampely JE, et al. Exercise therapy rescues skeletal muscle dysfunction and exercise intolerance in cardiometabolic HFpEF. *JACC Basic Transl Sci.* 2024;9:1409–1425.

170. García-Domínguez E, García-Domínguez C, Cabrera-Alarcón JL, et al. Mitochondrial remodeling in skeletal muscle underlies exercise-induced reversal of age-associated functional decline in mice and humans. *Proc Natl Acad Sci U S A.* 2026;123:e2508286123.

171. Joanisse S, Nederveen JP, Baker JM, Snijders T, Iacono C, Parise G. Exercise conditioning in old mice improves skeletal muscle regeneration. *FASEB J.* 2016;30:3256–3268.

172. Chen Z, Li L, Wu W, et al. Exercise protects proliferative muscle satellite cells against exhaustion via the Igfbp7-Akt-mTOR axis. *Theranostics.* 2020;10:6448–6466.

173. Winzer EB, Augstein A, Schauer A, et al. Impact of different training modalities on molecular alterations in skeletal muscle of patients with heart failure with preserved ejection fraction: a substudy of the OptimEx trial. *Circ Heart Fail.* 2022;15:e009124.

174. Masschelein E, D'Hulst G, Zvick J, et al. Exercise promotes satellite cell contribution to myofibers in a load-dependent manner. *Skeletal Muscle.* 2020;10:21.

175. Mangan G, Bombardier E, Mitchell AS, Quadrilatero J, Tiidus PM. Oestrogen-dependent satellite cell activation and proliferation following a running exercise occurs via the PI3K signalling pathway and not IGF-1. *Acta Physiol (Oxf).* 2014;212:75–85.

176. Vacca A, Wang R, Nambiar N, et al. Lifestyle interventions in cardiometabolic HFpEF: dietary and exercise modalities. *Heart Fail Rev.* 2025;30:637–655.

177. Das JK, Banskota N, Candia J, et al. Calorie restriction modulates the transcription of genes related to stress response and longevity in human muscle: the CALERIE study. *Aging Cell.* 2023;22:e13963.

178. El Hajj EC, El Hajj MC, Sykes B, et al. Pragmatic weight management program for patients with obesity and heart failure with preserved ejection fraction. *J Am Heart Assoc.* 2021;10:e022930.

179. Brubaker PH, Nicklas BJ, Houston DK, et al. A randomized, controlled trial of resistance training added to caloric restriction plus aerobic exercise training in obese heart failure with preserved ejection fraction. *Circ Heart Fail.* 2023;16:e010161.

180. Upadhyay B, Brubaker PH, Nicklas BJ, Houston DK, Haykowsky MJ, Kitzman DW. Long-term changes in body composition and exercise capacity following calorie restriction and exercise training in older patients with obesity and heart failure with preserved ejection fraction. *J Card Fail.* 2025;31:497–507.

181. Weiss EP, Racette SB, Villareal DT, et al. Lower extremity muscle size and strength and aerobic capacity decrease with caloric restriction but not with exercise-induced weight loss. *J Appl Physiol* (1985). 2007;102:634–640.
182. Hummel SL, Seymour EM, Brook RD, et al. Low-sodium DASH diet improves diastolic function and ventricular-arterial coupling in hypertensive heart failure with preserved ejection fraction. *Circ Heart Fail*. 2013;6:1165–1171.
183. Yeo D, Zhang T, Liu T, Zhang Y, Kang C, Ji LL. Protective effects of extra virgin olive oil and exercise training on rat skeletal muscle against high-fat diet feeding. *J Nutr Biochem*. 2022;100:108902.
184. Kamolrat T, Gray SR, Thivierge MC. Fish oil positively regulates anabolic signalling alongside an increase in whole-body gluconeogenesis in ageing skeletal muscle. *Eur J Nutr*. 2013;52:647–657.
185. Cornish SM, Cordingley DM, Shaw KA, et al. Effects of omega-3 supplementation alone and combined with resistance exercise on skeletal muscle in older adults: a systematic review and meta-analysis. *Nutrients*. 2022;14:2221.
186. Hearon CM, Dias KA, MacNamara JP, et al. 1 year HIIT and omega-3 fatty acids to improve cardiometabolic risk in stage-a heart failure. *JACC Heart Fail*. 2022;10:238–249.
187. Carbone S, Billingsley HE, Ahmed SI, et al. Unsaturated fatty acids to improve cardiorespiratory fitness in patients with obesity-related heart failure with preserved ejection fraction: the UFA-Preserved2 randomized controlled crossover study. *Circ Heart Fail*. 2026;19:e013511.
188. Alves PKN, Schauer A, Augstein A, et al. Leucine supplementation prevents the development of skeletal muscle dysfunction in a rat model of HFpEF. *Cells*. 2024;13:502.
189. Aquilani R, Viglio S, Iadarola P, et al. Oral amino acid supplements improve exercise capacities in elderly patients with chronic heart failure. *Am J Cardiol*. 2008;101:104E–110E.
190. Wu C, Kato TS, Ji R, et al. Supplementation of L-alanyl-L-glutamine and fish oil improves body composition and quality of life in patients with chronic heart failure. *Circ Heart Fail*. 2015;8:1077–1087.
191. Chaudhari M, Makhoulfi J, Doelling B, et al. Structural and metabolic remodeling of skeletal muscle in heart failure with reduced ejection fraction: a review: beyond the failing heart. *Int J Mol Sci*. 2026;27:2886.
192. del Favero S, Roschel H, Solis MY, et al. Beta-alanine (Carnosyn™) supplementation in elderly subjects (60–80 years): effects on muscle carnitine content and physical capacity. *Amino Acids*. 2012;43:49–56.
193. Lombardi C, Carubelli V, Lazzarini V, et al. Effects of oral administration of orodispersible levo-carnosine on quality of life and exercise performance in patients with chronic heart failure. *Nutrition*. 2015;31:72–78.
194. Challa AA, Hill BG, Nystoriak MA, Gouwens KR, Kalra DK. Ketone bodies in cardiovascular disease: the vasculature as a therapeutic target. *JACC Basic Transl Sci*. 2025;10:101328.
195. Selvaraj S, Karaj A, Chirinos JA, et al. Crossover trial of exogenous ketones on cardiometabolic endpoints in heart failure with preserved ejection fraction. *JACC Heart Fail*. 2025;13:102435.
196. Packer M, Zile MR, Kramer CM, et al. Tirzepatide for heart failure with preserved ejection fraction and obesity. *N Engl J Med*. 2025;392:427–437.
197. Kosiborod MN, Abildstrøm SZ, Borlaug BA, et al. Semaglutide in patients with heart failure with preserved ejection fraction and obesity. *N Engl J Med*. 2023;389:1069–1084.
198. Kosiborod MN, Petrie MC, Borlaug BA, et al. Semaglutide in patients with obesity-related heart failure and type 2 diabetes. *N Engl J Med*. 2024;390:1394–1407.
199. Borlaug BA, Kitzman DW, Davies MJ, et al. Semaglutide in HFpEF across obesity class and by body weight reduction: a prespecified analysis of the STEP-HFpEF trial. *Nat Med*. 2023;29:2358–2365.
200. Borlaug BA, Zile MR, Kramer CM, et al. Effects of tirzepatide on circulatory overload and end-organ damage in heart failure with preserved ejection fraction and obesity: a secondary analysis of the SUMMIT trial. *Nat Med*. 2025;31:544–551.
201. Capone F, Nambiar N, Schiattarella GG. Beyond weight loss: the emerging role of incretin-based treatments in cardiometabolic HFpEF. *Curr Opin Cardiol*. 2024;39:148–153.
202. Withaar C, Meems LMG, Nollet EE, et al. The cardioprotective effects of semaglutide exceed those of dietary weight loss in mice with HFpEF. *JACC Basic Transl Sci*. 2023;8:1298–1314.
203. Withaar C, Meems LMG, Markousis-Mavrogenis G, et al. The effects of liraglutide and dapagliflozin on cardiac function and structure in a multi-hit mouse model of heart failure with preserved ejection fraction. *Cardiovasc Res*. 2021;117:2108–2124.
204. Gurjar AA, Kushwaha S, Chattopadhyay S, et al. Long acting GLP-1 analog liraglutide ameliorates skeletal muscle atrophy in rodents. *Metabolism*. 2020;103:154044.
205. Ren Q, Chen S, Chen X, et al. An effective glucagon-like peptide-1 receptor agonists, semaglutide, improves sarcopenic obesity in obese mice by modulating skeletal muscle metabolism. *Drug Des Devel Ther*. 2022;16:3723–3735.
206. Hong Y, Lee JH, Jeong KW, Choi CS, Jun HS. Amelioration of muscle wasting by glucagon-like peptide-1 receptor agonist in muscle atrophy. *J Cachexia Sarcopenia Muscle*. 2019;10:903–918.
207. Li Z, Ni CL, Yao Z, Chen LM, Niu WY. Liraglutide enhances glucose transporter 4 translocation via regulation of AMP-activated protein kinase signaling pathways in mouse skeletal muscle cells. *Metabolism*. 2014;63:1022–1030.
208. Yamada S, Ogura Y, Inoue K, et al. Effect of GLP-1 receptor agonist, liraglutide, on muscle in spontaneously diabetic torii fatty rats. *Mol Cell Endocrinol*. 2022;539:111472.
209. von Haehling S, Sato R, Langer H, et al. Muscle loss in obesity therapy as a therapeutic target: trial design and endpoints for regulatory discussions. *J Cachexia Sarcopenia Muscle*. 2025;16:e70147.
210. Wilding JPH, Batterham RL, Calanna S, et al. Once-weekly semaglutide in adults with overweight or obesity. *N Engl J Med*. 2021;384:989–1002.
211. Jastreboff AM, Aronne LJ, Ahmad NN, et al. Tirzepatide once weekly for the treatment of obesity. *N Engl J Med*. 2022;387:205–216.
212. Coskun T, Wu Q, Schloot NC, et al. Effects of retatrutide on body composition in people with type 2 diabetes: a substudy of a phase 2, double-blind, parallel-group, placebo-controlled, randomised trial. *Lancet Diabetes Endocrinol*. 2025;13:674–684.
213. Sattar N, Neeland IJ, Dahlqvist Leinhard O, et al. Tirzepatide and muscle composition changes in people with type 2 diabetes (SURPASS-3 MRI): a post-hoc analysis of a randomised, open-label, parallel-group, phase 3 trial. *Lancet Diabetes Endocrinol*. 2025;13:482–493.
214. Alissou M, Demangeat T, Folope V, et al. Impact of semaglutide on fat mass, lean mass and muscle function in patients with obesity: the SEMALEAN study. *Diabetes Obes Metab*. 2026;28:112–121.
215. Tinsley GM, Heymsfield SB. Fundamental body composition principles provide context for fat-free and skeletal muscle loss with GLP-1 RA treatments. *J Endocr Soc*. 2024;8:bvae164.
216. Karasawa T, Choi RH, Meza CA, et al. Unexpected effects of semaglutide on skeletal muscle mass and force-generating capacity in mice. *Cell Metab*. 2025;37:1619–1620.
217. Xiang J, Qin L, Zhong J, Xia N, Liang Y. GLP-1RA liraglutide and semaglutide improves obesity-induced muscle atrophy via SIRT1 pathway. *Diabetes Metab Syndr Obes*. 2023;16:2433–2446.
218. Nunn E, Jaiswal N, Gavin M, et al. Antibody blockade of activin type II receptors preserves skeletal muscle mass and enhances fat loss during GLP-1 receptor agonism. *Mol Metab*. 2024;80:101880.
219. Langer HT, Gilmore NK, Hayden CMT, et al. Weight loss with GLP-1 medicines does not result in a disproportionate loss of muscle mass or function in obese mice and humans. *Cell Rep Med*. 2026;7:102665.
220. Nambu H, Takada S, Fukushima A, et al. Empagliflozin restores lowered exercise endurance capacity via the activation of skeletal muscle fatty acid oxidation in a murine model of heart failure. *Eur J Pharmacol*. 2020;866:172810.
221. Winzer EB, Schauer A, Langner E, et al. Empagliflozin preserves skeletal muscle function in a HFpEF rat model. *Int J Mol Sci*. 2022;23:10989.
222. Souza LM, Damatto FC, Brandão BB, et al. Impact of SGLT2i on cardiac remodeling and the soleus muscle of infarcted rats. *Antioxidants (Basel)*. 2025;14:647.
223. Wood N, Straw S, Cheng CW, et al. Sodium-glucose cotransporter 2 inhibitors influence skeletal muscle pathology in patients with heart

failure and reduced ejection fraction. *Eur J Heart Fail.* 2024;26:925-935.

224. Desai A, Sharma S, Abuah N, et al. Impact of sodium-glucose cotransporter-2 inhibitor use on peak VO. *Front Cardiovasc Med.* 2024;11:1376645.

225. Lewis GD, Gosch K, Cohen LP, et al. Effect of dapagliflozin on 6-minute walk distance in heart failure with preserved ejection fraction: preserved-HF. *Circ Heart Fail.* 2023;16:e010633.

226. McMurray JJV, Docherty KF, de Boer RA, et al. Effect of dapagliflozin versus placebo on symptoms and 6-minute walk distance in patients with heart failure: the DETERMINE randomized clinical trials. *Circulation.* 2024;149:825-838.

227. Katano S, Yano T, Shimizu M, et al. Does renin-angiotensin system inhibition have impacts on muscle mass and bone mineral density in heart failure patients? *ESC Heart Fail.* 2021;8:2617-2624.

228. Lafont R, Serova M, Didry-Barca B, et al. 20-Hydroxyecdysone activates the protective arm of the RAAS via the MAS receptor. *J Mol Endocrinol.* 2021;68:77-87.

229. Cisternas F, Morales MG, Meneses C, et al. Angiotensin-(1-7) decreases skeletal muscle atrophy induced by angiotensin II through a Mas receptor-dependent mechanism. *Clin Sci (Lond).* 2015;128:307-319.

230. Serova M, Didry-Barca B, Deloux R, et al. BIO101 stimulates myoblast differentiation and improves muscle function in adult and old mice. *J Cachexia Sarcopenia Muscle.* 2024;15:55-66.

231. Fielding RA, Dao MM, Cannon K, et al. BIO101 in sarcopenic seniors at risk of mobility disability: results of a double-blind randomised interventional phase 2b trial. *J Cachexia Sarcopenia Muscle.* 2025;16:e13750.

232. Subbotina E, Sierra A, Zhu Z, et al. Musclin is an activity-stimulated myokine that enhances physical endurance. *Proc Natl Acad Sci U S A.* 2015;112:16042-16047.

233. Engeli S, Birkenfeld AL, Badin PM, et al. Natriuretic peptides enhance the oxidative capacity of human skeletal muscle. *J Clin Invest.* 2012;122:4675-4679.

234. Saw EL, Werner LD, Cooper HL, et al. Musclin counteracts skeletal muscle dysfunction and exercise intolerance in heart failure with preserved ejection fraction. *Circ Heart Fail.* 2025;18:e012350.

235. Reddy YNV, Stewart GM, Obokata M, Koeppe KE, Borlaug BA. Peripheral and pulmonary effects of inorganic nitrite during exercise in heart failure with preserved ejection fraction. *Eur J Heart Fail.* 2021;23:814-823.

236. Eggebeen J, Kim-Shapiro DB, Haykowsky M, et al. One week of daily dosing with beetroot juice improves submaximal endurance and blood pressure in older patients with heart failure and preserved ejection fraction. *JACC Heart Fail.* 2016;4:428-437.

237. Coggan AR, Leibowitz JL, Speare CA, et al. Acute dietary nitrate intake improves muscle contractile function in patients with heart failure: a double-blind, placebo-controlled, randomized trial. *Circ Heart Fail.* 2015;8:914-920.

238. Coggan AR, Broadstreet SR, Mahmood K, et al. Dietary nitrate increases VO. *J Card Fail.* 2018;24:65-73.

239. Zamani P, Shah SJ, Cohen JB, et al. Potassium nitrate in heart failure with preserved ejection fraction: a randomized clinical trial. *JAMA Cardiol.* 2025;10:284-289.

240. Coggan AR, Park LK, Racette SB, et al. The inorganic Nitrate and eXercise performance in heart failure (iNIX-HF) phase II clinical trial: rationale and study design. *Contemp Clin Trials Commun.* 2023;36:101208.

241. Armstrong PW, Pieske B, Anstrom KJ, et al. Vericiguat in patients with heart failure and reduced ejection fraction. *N Engl J Med.* 2020;382:1883-1893.

242. Zhan Y, Li L, Zhou J, et al. Efficacy of vericiguat in patients with chronic heart failure and reduced ejection fraction: a prospective observational study. *BMC Cardiovasc Disord.* 2025;25:83.

243. Li YL, Hu BA, Xiao P, et al. Soluble guanylate cyclase stimulator vericiguat alleviates skeletal muscle injury and exercise capacity decline in ApoE. *Biochem Pharmacol.* 2025;245:117570.

244. Armstrong PW, Lam CSP, Anstrom KJ, et al. Effect of vericiguat vs placebo on quality of life in patients with heart failure and preserved ejection fraction: the VITALITY-HFpEF randomized clinical trial. *JAMA.* 2020;324:1512-1521.

245. Di Lodovico E, Facondo P, Delbarba A, et al. Testosterone, hypogonadism, and heart failure. *Circ Heart Fail.* 2022;15:e008755.

246. Caminiti G, Volterrani M, Iellamo F, et al. Effect of long-acting testosterone treatment on functional exercise capacity, skeletal muscle performance, insulin resistance, and baroreflex sensitivity in elderly patients with chronic heart failure: a double-blind, placebo-controlled, randomized study. *J Am Coll Cardiol.* 2009;54:919-927.

247. Lincoff AM, Bhasin S, Flevaris P, et al. Cardiovascular safety of testosterone-replacement therapy. *N Engl J Med.* 2023;389:107-117.

248. Dalton JT, Barnette KG, Bohl CE, et al. The selective androgen receptor modulator GTx-024 (enobosarm) improves lean body mass and physical function in healthy elderly men and postmenopausal women: results of a double-blind, placebo-controlled phase II trial. *J Cachexia Sarcopenia Muscle.* 2011;2:153-161.

249. Dobs AS, Boccia RV, Croot CC, et al. Effects of enobosarm on muscle wasting and physical function in patients with cancer: a double-blind, randomized controlled phase 2 trial. *Lancet Oncol.* 2013;14:335-345.

250. Gao W, Reiser PJ, Coss CC, et al. Selective androgen receptor modulator treatment improves muscle strength and body composition and prevents bone loss in orchidectomized rats. *Endocrinology.* 2005;146:4887-4897.

251. Theodorakis N, Feretakis G, Kreouzi M, et al. GH therapy in chronic heart failure: a systematic review and meta-analysis of randomized controlled trials. *J Clin Endocrinol Metab.* 2025;110:e1252-e1260.

252. Marra AM, D'Assante R, De Luca M, et al. Growth hormone replacement therapy in heart

failure with reduced ejection fraction: a randomized, double-blind, placebo-controlled trial. *JACC Heart Fail.* 2025;13:602-614.

253. Temel JS, Abernethy AP, Currow DC, et al. Anamorelin in patients with non-small-cell lung cancer and cachexia (ROMANA 1 and ROMANA 2): results from two randomised, double-blind, phase 3 trials. *Lancet Oncol.* 2016;17:519-531.

254. Barazzoni R, Gortan Cappellari G, Palus S, et al. Acylated ghrelin treatment normalizes skeletal muscle mitochondrial oxidative capacity and AKT phosphorylation in rat chronic heart failure. *J Cachexia Sarcopenia Muscle.* 2017;8:991-998.

255. Palus S, Schur R, Akashi YJ, et al. Ghrelin and its analogues, BIM-28131 and BIM-28125, improve body weight and regulate the expression of MuRF-1 and MAFbx in a rat heart failure model. *PLoS One.* 2011;6:e26865.

256. Nagaya N, Moriya J, Yasumura Y, et al. Effects of ghrelin administration on left ventricular function, exercise capacity, and muscle wasting in patients with chronic heart failure. *Circulation.* 2004;110:3674-3679.

257. Pötsch MS, Tschirner A, Palus S, et al. The anabolic catabolic transforming agent (ACTA) espidolol increases muscle mass and decreases fat mass in old rats. *J Cachexia Sarcopenia Muscle.* 2014;5:149-158.

258. Stewart Coats AJ, Ho GF, Prabhaskar K, et al. Espindolol for the treatment and prevention of cachexia in patients with stage III/IV non-small cell lung cancer or colorectal cancer: a randomized, double-blind, placebo-controlled, international multicentre phase II study (the ACT-ONE trial). *J Cachexia Sarcopenia Muscle.* 2016;7:355-365.

259. Lach-Trifileff E, Minetti GC, Sheppard K, et al. An antibody blocking activin type II receptors induces strong skeletal muscle hypertrophy and protects from atrophy. *Mol Cell Biol.* 2014;34:606-618.

260. Heymsfield SB, Coleman LA, Miller R, et al. Effect of bimagrumab vs placebo on body fat mass among adults with type 2 diabetes and obesity: a phase 2 randomized clinical trial. *JAMA Netw Open.* 2021;4:e2033457.

261. Heymsfield SB, Aronne LJ, Montgomery P, et al. Bimagrumab plus semaglutide alone or in combination for the treatment of obesity: a randomized phase 2 trial. *Nat Med.* 2026;32:869-882.

262. Crawford TO, Darras BT, Day JW, et al. Safety and efficacy of apitegromab in patients with spinal muscular atrophy types 2 and 3: the phase 2 TOPAZ study. *Neurology.* 2024;102:e209151.

263. Pratley RE, Denham DS, Trivedi R, et al. Apitegromab for lean mass preservation during tirzepatide-induced weight loss: a randomized, double-blind, placebo-controlled phase 2 trial. *Nat Med.* 2026;32:2673-2678.

264. Servais L, Lair LL, Connolly AM, et al. Taldefgrobep alfa and the phase 3 RESILIENT trial in spinal muscular atrophy. *Int J Mol Sci.* 2024;25:10273.

265. Mastaitis JW, Gomez D, Raya JG, et al. GDF8 and activin A blockade protects against GLP-1-induced muscle loss while enhancing fat loss in obese male mice and non-human primates. *Nat Commun*. 2025;16:4377.
266. Mortensen SA, Rosenfeldt F, Kumar A, et al. The effect of coenzyme Q10 on morbidity and mortality in chronic heart failure: results from Q-SYMBIO: a randomized double-blind trial. *JACC Heart Fail*. 2014;2:641-649.
267. Al Saadi T, Assaf Y, Farwati M, et al. Coenzyme Q10 for heart failure. *Cochrane Database Syst Rev*. 2021;2:CD008684.
268. Rossman MJ, Santos-Parker JR, Steward CAC, et al. Chronic supplementation with a mitochondrial antioxidant (MitoQ) improves vascular function in healthy older adults. *Hypertension*. 2018;71:1056-1063.
269. Rostamzadeh F, Najafipour H, Aminzadeh S, Jafari E. Therapeutic effects of the combination of moderate-intensity endurance training and MitoQ supplementation in rats with isoproterenol-induced myocardial injury: the role of mitochondrial fusion, fission, and mitophagy. *Biomed Pharmacother*. 2024;170:116020.
270. Halliday BP, Owen R, Ragavan A, et al. A double-blind, randomized placebo-controlled trial examining the effect of MitoQ on myocardial energetics in patients with dilated cardiomyopathy. *Eur Heart J Cardiovasc Imaging*. 2026;27:74-77.
271. Pandey A, Lewis GD, Borlaug BA, et al. Novel controlled metabolic accelerator for obesity-related HFpEF: the HuMAIN-HFpEF randomized clinical trial. *JAMA Cardiol*. 2025;10:609-616.
272. Birk AV, Liu S, Soong Y, et al. The mitochondrial-targeted compound SS-31 re-energizes ischemic mitochondria by interacting with cardiolipin. *J Am Soc Nephrol*. 2013;24:1250-1261.
273. Sabbah HN, Gupta RC, Singh-Gupta V, Zhang K. Effects of elamipretide on skeletal muscle in dogs with experimentally induced heart failure. *ESC Heart Fail*. 2019;6:328-335.
274. Butler J, Khan MS, Anker SD, et al. Effects of elamipretide on left ventricular function in patients with heart failure with reduced ejection fraction: the PROGRESS-HF phase 2 trial. *J Card Fail*. 2020;26:429-437.
275. Liu S, D'Amico D, Shankland E, et al. Effect of urolithin A supplementation on muscle endurance and mitochondrial health in older adults: a randomized clinical trial. *JAMA Netw Open*. 2022;5:e2144279.
276. Singh A, D'Amico D, Andreux PA, et al. Urolithin A improves muscle strength, exercise performance, and biomarkers of mitochondrial health in a randomized trial in middle-aged adults. *Cell Rep Med*. 2022;3:100633.
277. Van Tassell BW, Canada J, Carbone S, et al. Interleukin-1 blockade in recently decompensated systolic heart failure: results from REDHART (recently decompensated heart failure anakinra response trial). *Circ Heart Fail*. 2017;10:e004373.
278. Van Tassell BW, Golino M, Canada JM, et al. Resolution of systemic inflammation in patients with recently decompensated heart failure with reduced ejection fraction with and without Interleukin-1 blockade by anakinra. *Circ Heart Fail*. 2026;19:e013546.
279. Taylor EB, Hall JE, Mouton AJ. Current anti-inflammatory strategies for treatment of heart failure: from innate to adaptive immunity. *Pharmacol Res*. 2025;216:107761.
280. Groarke JD, Crawford J, Collins SM, et al. Ponesemromab for the treatment of cancer cachexia. *N Engl J Med*. 2024;391:2291-2303.
281. Narkar VA, Downes M, Yu RT, et al. AMPK and PPARdelta agonists are exercise mimetics. *Cell*. 2008;134:405-415.
282. Steinberg GR, Carling D. AMP-activated protein kinase: the current landscape for drug development. *Nat Rev Drug Discov*. 2019;18:527-551.
283. Misquitta NS, Ravel-Chapuis A, Jasmin BJ. Combinatorial treatment with exercise and AICAR potentiates the rescue of myotonic dystrophy type 1 mouse muscles in a sex-specific manner. *Hum Mol Genet*. 2023;32:551-566.
284. Marcinko K, Bujak AL, Lally JS, et al. The AMPK activator R419 improves exercise capacity and skeletal muscle insulin sensitivity in obese mice. *Mol Metab*. 2015;4:643-651.
285. Montalvo RN, Li X, Many GM, et al. Ampk alpha2 T172 activation dictates exercise performance and energy transduction in skeletal muscle. *Sci Adv*. 2026;12:eab3338.
286. Ericsson M, Steneberg P, Nyrén R, Edlund H. AMPK activator O304 improves metabolic and cardiac function, and exercise capacity in aged mice. *Commun Biol*. 2021;4:1306.
287. Larsen AH, Jessen N, Nørrelund H, et al. A randomised, double-blind, placebo-controlled trial of metformin on myocardial efficiency in insulin-resistant chronic heart failure patients without diabetes. *Eur J Heart Fail*. 2020;22:1628-1637.
288. Sung MM, Byrne NJ, Robertson IM, et al. Resveratrol improves exercise performance and skeletal muscle oxidative capacity in heart failure. *Am J Physiol Heart Circ Physiol*. 2017;312:H842-H853.
289. Gal R, Deres L, Horvath O, et al. Resveratrol improves heart function by moderating inflammatory processes in patients with systolic heart failure. *Antioxidants (Basel)*. 2020;9:1108.
290. Steneberg P, Lindahl E, Dahl U, et al. PAN-AMPK activator O304 improves glucose homeostasis and microvascular perfusion in mice and type 2 diabetes patients. *JCI Insight*. 2018;3:e99114.
291. Matsumoto J, Takada S, Furihata T, et al. Brain-derived neurotrophic factor improves impaired fatty acid oxidation via the activation of adenosine monophosphate-activated protein Kinase- α - proliferator-activated receptor- γ coactivator-1 α signaling in skeletal muscle of mice with heart failure. *Circ Heart Fail*. 2021;14:e005890.
292. Matsumoto J, Takada S, Kinugawa S, et al. Brain-derived neurotrophic factor improves limited exercise capacity in mice with heart failure. *Circulation*. 2018;138:2064-2066.
293. Iwabu M, Okada-Iwabu M, Tanabe H, et al. AdipoR agonist increases insulin sensitivity and exercise endurance in AdipoR-humanized mice. *Commun Biol*. 2021;4:45.
294. Balasubramanian P, Schaar AE, Gustafson GE, et al. Adiponectin receptor agonist AdipoRon improves skeletal muscle function in aged mice. *Elife*. 2022;11:e71282.
295. Selvais CM, Davis-López de Carrizosa MA, Nacht M, et al. AdipoRon enhances healthspan in middle-aged obese mice: striking alleviation of myosteatosis and muscle degenerative markers. *J Cachexia Sarcopenia Muscle*. 2023;14:464-478.
296. Bekfani T, Bekhite M, Neugebauer S, et al. Metabolomic profiling in patients with heart failure and exercise intolerance: kynurenine as a potential biomarker. *Cells*. 2022;11:1674.
297. Ramirez MF, Lau ES, Parekh JK, et al. Obesity-related biomarkers are associated with exercise intolerance and HFpEF. *Circ Heart Fail*. 2023;16:e010618.
298. Tang G, Du Y, Guan H, et al. Butyrate ameliorates skeletal muscle atrophy in diabetic nephropathy by enhancing gut barrier function and FFA2-mediated PI3K/Akt/mTOR signals. *Br J Pharmacol*. 2022;179:159-178.
299. Agrawal A, Rathor R, Kumar R, Singh SN, Kumar B, Suryakumar G. Endogenous dipeptide-carnosine supplementation ameliorates hypobaric hypoxia-induced skeletal muscle loss via attenuating endoplasmic reticulum stress response and maintaining proteostasis. *IUBMB Life*. 2022;74:101-116.
300. Zizola C, Kennel PJ, Akashi H, et al. Activation of PPAR δ signaling improves skeletal muscle oxidative metabolism and endurance function in an animal model of ischemic left ventricular dysfunction. *Am J Physiol Heart Circ Physiol*. 2015;308:H1078-H1085.
301. Laitano O, Ahn B, Patel N, et al. Pharmacological targeting of mitochondrial reactive oxygen species counteracts diaphragm weakness in chronic heart failure. *J Appl Physiol (1985)*. 2016;120:733-742.
302. Adams V, Schauer A, Augstein A, et al. Targeting MuRF1 by small molecules in a HFpEF rat model improves myocardial diastolic function and skeletal muscle contractility. *J Cachexia Sarcopenia Muscle*. 2022;13:1565-1581.
303. Zhang W, You B, Qi D, et al. Trimetazidine and exercise provide comparable improvements to high fat diet-induced muscle dysfunction through enhancement of mitochondrial quality control. *Sci Rep*. 2021;11:19116.

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